

SOCIAL MEDIA ADVERTISING LOADS AS PRICES^{*}

George Beknazar-Yuzbashev[†] Rafael Jiménez-Durán[‡]
Andrey Simonov[§] Mateusz Stalinski[¶]

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Abstract

Most digital platforms are funded through advertising rather than direct payments. Why? We argue that three main factors could explain this prevalence: users are more sensitive to monetary prices than to ad loads, microtargeting improves the match quality between users and ads, and platforms can personalize ad loads and thus price discriminate. We conduct a field experiment on Facebook with 1,810 users who install a browser extension that (i) hides nearly all ads or (ii) replaces microtargeted ads with untargeted ones. We find that hiding 82% of ads increases time on the platform by only 6%, showing that users are highly insensitive to ad loads. Removing targeting sharply reduces ad clicks and long-run engagement, indicating that targeting increases the match quality between users and ads. Finally, two-thirds of ad-load variation occurs across users, consistent with ad-load discrimination. Counterfactual simulations indicate that an ad-funded model performs at least as well as a subscription model in terms of profits and delivers higher consumer surplus. The key mechanism is that users are much less sensitive to ad loads than to monetary prices, making advertising a relatively efficient revenue source.

Keywords: social media platforms, online advertisement, user engagement, field experiment

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[†]University of Chicago, beknazar@uchicago.edu.

[‡]Bocconi University, IGIER, Stigler Center, CESifo, and CEPR, rafael.jimenez@unibocconi.it.

[§]Columbia University and CEPR, as5443@gsb.columbia.edu.

[¶]University of Warwick and CAGE, mateusz.stalinski@warwick.ac.uk.

1 Introduction

Advertising-funded social media platforms have been criticized for maximizing engagement at the expense of content quality, contributing to the spread of misinformation, clickbait, and divisive content (Liu et al. 2022; Acemoglu et al. 2024; Beknazar-Yuzbashev et al. 2024). Subscriptions are often proposed as a remedy, yet advertising has dominated since the earliest social platforms (e.g., SixDegrees.com in 1997) and still accounts for over 97% of Meta’s revenue—even as technology has made direct payments viable.¹ Why does the ad-funded model remain so prevalent? This question has become increasingly urgent as regulations—such as the EU’s Digital Markets Act and Digital Services Act—and privacy-related product changes constrain platforms’ ability to advertise and microtarget.

This paper investigates why advertising dominates social media monetization and what happens when this model is restricted. Treating ads as implicit prices that users pay in attention rather than money (Anderson and Coate 2005; Calvano and Polo 2021), we suggest three channels that may explain advertising’s prevalence. First, users may be less sensitive to attention prices than to monetary prices, making advertising a relatively efficient way to extract surplus. Second, microtargeting may improve the match between users and ads—a channel that depends on data practices increasingly subject to privacy regulation. Third, platforms can personalize ad loads across users, a form of price discrimination that may be harder to implement with monetary prices.

We first develop a modeling framework to formalize these channels. Users allocate time to a platform that monetizes through advertising, and the platform can personalize both ad loads and ad targeting based on observable user characteristics. The model shows that optimal ad loads follow an inverse-elasticity logic: users less sensitive to ads face higher loads. Network effects also matter: users whose participation generates larger spillovers receive lower ad loads, as the platform protects their engagement. Finally, the framework delivers a decomposition comparing ad-based and subscription-based profits, isolating the contributions of microtargeting, ad-load personalization, and differential demand sensitivity to attention versus monetary prices.

To estimate the key model parameters, we conduct a field experiment with 1,810 U.S. Facebook users who install a custom desktop browser extension. The extension passively

¹See <https://www.sec.gov/Archives/edgar/data/1326801/000132680125000017/meta-20241231.htm>, accessed February 04, 2026.

collects platform activity and browsing data for six weeks. Users are then randomly assigned to one of two interventions for six more weeks. In the first treatment arm, the extension hides most ads, reducing ad load to near zero. In the second arm, ads appear at the same frequency but are replaced with ads originally shown to other users, degrading match quality while holding quantity fixed. Each intervention has a dedicated control group, with the *Replace Control* replacing the ad with itself.² Comparing outcomes across treatment and control groups before and after randomization identifies the causal effects of changing ad load and ad targeting.

We measure four categories of outcomes: ad and platform engagement on Facebook, time spent on competing platforms, and platform valuation. Ad engagement is measured through ad impressions, ad clicks, and a summary index of the two; platform engagement is measured through active time, the volume of posts and comments consumed, and an analogous index. To capture potential substitution, we track time spent on competing platforms, including X and Reddit. For platform valuation, we elicit users’ willingness to accept (WTA) to deactivate Facebook for four weeks (Allcott et al. 2020), surveyed at both baseline and endline. The endline survey was administered at the end of the six-week intervention period. Subsequently, the extension continued the intervention for approximately thirteen additional weeks, allowing us to assess whether treatment effects persist.

Our first result is that social media feed ad loads—the ratio of ad content to total platform content in the main feed—are highly heterogeneous across users even in the absence of intervention, consistent with systematic ad-load discrimination. Averaged at the user level, untreated feed ad loads on Facebook were approximately 13% (roughly one ad per eight pieces of content). We document substantial underlying heterogeneity: the cross-user standard deviation is 4.6 percentage points, roughly one-third of the mean. Indeed, 67% of the daily variation in ad load on Facebook is attributable to individual fixed effects, suggesting that platforms systematically charge different users different “attention prices.” This cross-user variation correlates with observable demographics—most notably age, desktop usage share, and political affiliation. To assess whether ad-load discrimination is specific to Facebook or a broader feature of social media monetization, we also examine ad loads on X and Reddit, finding similar averages (11% and 9%) and similar shares of variation explained by individual

²This design isolates the effect of targeting from the potential impact of the replacement arm’s technical implementation. For example, because the possibility to comment was disabled on replaced ads, comparisons between *Replace* and *Hide Control*—where commenting remained enabled—could confound targeting changes with functionality changes. *Replace Control* is designed to avoid such contamination. See Section 3 for details.

fixed effects (70% and 72%). Moreover, individual ad loads are positively correlated across Facebook, X, and Reddit, consistent with different platforms tailoring ad loads to similar user characteristics.

Turning to the experiment, we find that users are highly inelastic with respect to ad loads. The hiding intervention reduced experienced ad loads by 82%, yet content consumption increased by only 9% and active time by only 6% over the six-week intervention period. The preregistered platform engagement index rose by just 0.03 standard deviations (SD). These modest responses imply an elasticity below 0.1. Moreover, the effects remain stable over nineteen weeks of continued intervention (0.02 SD), suggesting muted long-run responses.

Our replacement intervention decreased ad targeting by substituting users’ algorithmically-selected ads with ads originally shown to other users: a measure of distance between user characteristics and the ads displayed to them increased by 0.5 SD. This intervention reduced ad engagement: the index fell by 0.05 SD, driven primarily by a 0.25 SD decline in ad clicks. Even more strikingly, the intervention reduced platform engagement as well—in the long run, active time and content consumption both fell by 13%, and the platform engagement index declined by 0.03 SD. Thus, degrading ad targeting decreases both ad engagement and platform engagement.

We report several supplementary results and robustness checks. First, Facebook’s algorithm showed no response to the hiding intervention—ad loads supplied by the platform remained unchanged—but responded mildly to the replacement intervention by supplying less-targeted ads. This asymmetry suggests the platform treats users’ ad-load sensitivity as stable while assuming that preferences for ad content can evolve. Second, we find little evidence of substitution toward competing platforms, consistent with Aridor (2025)—even large changes in ad load or targeting did not drive users to competitors. Third, neither intervention affected willingness to accept payment to deactivate Facebook, our proxy for platform valuation. Finally, our results are robust to differential attrition: survival at the end of the six-week intervention period exceeds 75% across all treatment arms and does not vary by assignment. Attrition remains non-differential after nineteen weeks of continued intervention, with survival above 50% in all groups.

Next, we use our experimental estimates to simulate counterfactual monetization regimes. We compare outcomes under three scenarios: the observed advertising-based model with personalized ad loads and microtargeting, a subscription-based model with uniform monetary pricing and no ads, and a hybrid “opt-in” alternative where users choose between an ad-

supported experience and a paid ad-free tier. For each regime, we evaluate profits, user engagement, platform demand, and consumer surplus. To identify which mechanisms drive the differences across regimes, we decompose performance gaps into four components: (i) the loss of ad-user match quality from eliminating ad targeting, (ii) the loss of ad price differentiation that ad targeting enables on the advertiser side, (iii) the loss of ad-load personalization across users, and (iv) the direct effect of switching from attention-based to monetary pricing.

The simulations indicate that the ad-funded model generates profits comparable to a subscription-based alternative while delivering higher engagement and consumer surplus. However, the mechanisms differ across outcomes. For profits, the main advantage of advertising comes from ad price differentiation—the ability to charge advertisers different rates for access to different user segments. For consumer surplus and platform demand, the differences are driven primarily by asymmetric demand sensitivity: users respond far more strongly to monetary prices than to advertising intensity. Even the hybrid opt-in model—similar to the paid ad-free tier Meta introduced in Europe under the Digital Markets Act—fails to outperform pure advertising: while around 17% of users in our simulations choose the subscription option, the aggregate outcomes are largely unchanged.

Taken together, these results help explain why advertising has remained the dominant revenue model for social media platforms despite the technological feasibility of alternatives. The persistence of this business model does not appear to reflect consumer exploitation; rather, advertising functions as an efficient pricing mechanism when users are highly sensitive to monetary charges but tolerant of attention costs. These findings paint a nuanced picture of advertising’s welfare effects: while previous research has highlighted that engagement-maximization incentives can inadvertently lead platforms to prioritize “toxic” content (Beknazar-Yuzbashev et al. 2024; Beknazar-Yuzbashev et al. 2025), this paper shows that ad-based monetization is nonetheless a relatively efficient revenue source both for platforms and users—holding constant the quality of content on the platform.

Our paper builds on the theoretical insight that advertising intensity functions as an implicit “attention price” (e.g., Anderson and Coate 2005; Anderson and De Palma 2012; Anderson and Peitz 2023; Calvano and Polo 2020; see Calvano and Polo 2021 for a review). Empirically, a negative relationship between ad loads and consumption has been documented in traditional media (Wilbur 2008) and in digital settings such as sponsored search, streaming audio, and e-commerce (Moshary 2021; Goli et al. 2025a; Goli et al. 2025b). In the context of social media, Brynjolfsson et al. (2025) report results from a 9-year long field ex-

periment on Facebook and find an elasticity of time spent with respect to ad loads of 0.094, consistent with our finding of modest engagement responses.³ We extend this literature in two ways: by experimentally varying not only ad quantity but also ad-user match quality, and by documenting systematic ad-load personalization across users consistent with price discrimination.

We also contribute to the growing literature on the economics of social media and platform monetization (Allcott et al. 2020; Aridor et al. 2024; Aridor et al. 2025b). This work highlights the impact of ad-driven incentives on the proliferation of harmful content (Liu et al. 2022; Acemoglu et al. 2024; Beknazar-Yuzbashev et al. 2024). Ad load policies are central to competition-policy and antitrust questions related to social media (e.g. see Ambrus et al. 2016; Athey et al. 2018; Center 2019; Argentesi et al. 2021; Aridor 2025). The most closely related paper is Katz and Allcott (2025), who develop a two-sided market model of digital media mergers that centers on advertiser demand, duplication inefficiencies, and equilibrium ad prices. Compared to Katz and Allcott (2025), we focus on the consumer side of the market, estimating user ad-load sensitivity and the engagement effects of microtargeting. Our counterfactual analysis of alternative ad load designs and monetization regimes speaks directly to ongoing regulatory scrutiny, including the European Commission’s investigation of Meta under the Digital Markets Act (European Commission 2025). Our results help to rationalize why advertising-based business models are so prevalent in the context of digital platforms.

Lastly, we contribute to the literature on the economics of privacy (Lin 2022; Goldfarb and Que 2023; Dubé et al. 2025). Prior work has shown that limiting tracking reduces advertising effectiveness and platform revenues (Goldfarb and Tucker 2011; Alcobendas et al. 2023; Aridor et al. 2023; Johnson 2024; Aridor et al. 2025a; Wernerfelt et al. 2025), but these estimates combine effects on both sides of the market. We isolate the consumer response by experimentally degrading ad targeting while holding the advertiser side fixed, finding that worse targeting reduces user engagement but does not affect platform valuation. This is consistent with the privacy paradox (Acquisti and Grossklags 2005): while users express growing privacy concerns across ever-broader contexts (Goldfarb and Tucker 2012), their revealed behavior suggests limited welfare costs from targeting.

³Relatedly, Goodman et al. (2026) apply a Beckerian time-allocation model to the experiment in Brynjolfsson et al. (2025) and an additional experiment, and show that Facebook and Instagram demand is highly inelastic to ad load and that time costs and time shares shape diversion patterns across online and offline activities.

The paper proceeds as follows. Section 2 presents a conceptual framework. Section 3 presents our experimental design. Section 4 presents our results. Section 5 evaluates counterfactual business models and mechanisms. Section 6 concludes.

2 Model

We now present a model where users spend time on a platform. The platform can personalize both the quantity of ads it serves to the users and their average match quality, and can in principle charge users a monetary subscription fee. We use this model 1) to generate predictions of how a profit-maximizing platform would optimally set individual ad loads and 2) to understand the conditions that determine the relative profitability of an ad-based platform relative to subscription-based platform. We consider these two extremes as a starting point.

Users. There is a set of individuals who choose how much time to spend on the platform in order to maximize their utility. Individuals derive utility from joining the platform, from the time they spend consuming content, and from the ads they are exposed to. They have observable types $\theta \in \Theta$, distributed according to the discrete probability mass function g^θ . We interpret θ as the set of characteristics that the platform can observe and condition on when personalizing ad loads or targeting. Within each observable type θ , there is a continuum of users indexed by an idiosyncratic type ε , which determines the “membership benefits” of joining the platform (and becoming a user). This idiosyncratic type is distributed according to the conditional density $f_{\varepsilon|\theta}$.

Consider an individual of type θ who allocates time $t^\theta \geq 0$ to consume content, conditional on joining the platform. The individual takes as given the (potentially personalized) ad load denoted by $a^\theta \in [0, 1]$, which denotes the amount of time spent consuming ads for each unit of time spent consuming content. Thus, the total amount of ads consumed is $A^\theta = a^\theta t^\theta$. The individual also takes as given the average ad-user match quality denoted by $q^\theta \in [0, 1]$. The total amount of time spent on the platform, conditional on joining, is $t^\theta(1 + a^\theta)$. Each unit of time spent carries an opportunity cost w^θ , reflecting either the value of time or the implicit cost of sharing personal data through engagement.

Conditional on joining the platform, the user’s optimization problem, considering the

utility from content consumption, ad consumption, and the opportunity cost of time, is:

$$v^\theta(a^\theta, q^\theta, x) = \max_{t \geq 0} u^{\theta,c}(t) + u^{\theta,a}(ta^\theta, q^\theta) + u^{\theta,j}(x) - w^\theta t(1 + a^\theta),$$

cross θ .
budget constraint ?

where $u^{\theta,c}$ is the utility of content consumption. We assume that the marginal utility of time spent is positive, $u_t^{\theta,c} > 0$. The utility from consuming ads is $u^{\theta,a}$, which depends both on the quantity and match quality of ads consumed. The marginal utility from ads, represented by $u_A^{\theta,a}$, can be positive or negative, but we restrict it to be lower than the marginal utility of engagement: $u_A^{\theta,a} < u_t^{\theta,c}$ in other words, users prefer to spend time consuming content vs. consuming ads. The match quality of ads has a “vertical” nature, in the sense that it increases the marginal utility of ads, $u_{Aq}^{\theta,a} \geq 0$. However, users can still derive positive or negative utility from consuming ads with a better match quality; for example, reflecting that they might have some instrumental utility from privacy (Lin 2022), which would result in users experiencing a negative marginal utility from a better match quality.⁴ The function $u^{\theta,j}(x)$ depends on the platform’s market share, x , to capture direct network effects from other users’ participation. Lastly, to ensure that second-order conditions hold, we assume that utilities are concave in t . We denote by $t^\theta(a^\theta, q^\theta)$ the solution to this optimization problem.

Let $a = (a^\theta)_{\theta \in \Theta}$ and $q = (q^\theta)_{\theta \in \Theta}$ denote the vectors of ad loads and qualities for all user types, respectively. Additionally, let s denote the “subscription” (monetary price) charged by the platform.⁵ The demand for the platform of θ users is given by $X^\theta(a^\theta, q^\theta, s, x) = g^\theta \int_{s-v^\theta(a^\theta, q^\theta, x)}^\infty f_{\varepsilon|\theta}(\varepsilon | \theta) d\varepsilon$, and total platform demand is $X(a, q, s, x) = \sum_\theta X^\theta(a^\theta, q^\theta, s, x)$. In equilibrium, this demand has to be consistent with expectations, $x = X(a, q, s, x)$. As standard in the network-effects literature, we assume that network effects are not too strong, such that, for any (a, q, s) , there exists a unique solution $x(a, q, s)$ to the fixed-point problem.

Platform. Let p^θ denote the price (in dollars per unit of time) that advertisers pay per unit of engagement of θ users with their ads, and p denote the vector of ad prices for all

⁴We normalize $u^{\theta,a}(0, q) = 0$ for all q : when there are no ads, their match quality is irrelevant.

⁵We assume that the platform cannot price-discriminate through subscription fees, but can personalize ad loads. This reflects common practice in real-world platforms.

users. The platform's profit function is then:

$$\pi(p, a, q, s) = \sum_{\theta} \left[\underbrace{p^{\theta} \times a^{\theta} \times t^{\theta}(a^{\theta}, q^{\theta}) \times x^{\theta}(a, q, s)}_{\text{Ad revenue}} + \underbrace{s \times x^{\theta}(a, q, s)}_{\text{Subscription revenue}} \right],$$

$p^{\theta}(a, q)$

where $x^{\theta}(a, q, s) = X^{\theta}(a^{\theta}, q^{\theta}, s, x(a, q, s))$ is the equilibrium demand of θ users. We can allow ad prices p^{θ} to depend on the ad load and match quality vectors, to capture potential information overload (Anderson and De Palma 2012), that advertiser willingness to pay increases with a better match quality, or that the platform has market power in digital ads markets (Gentzkow et al. 2024).

The optimization problem of an ad-based platform is then:

$$\max_{a, q} \pi(p(a, q), a, q, s = 0),$$

with solution a^*, q^* and a maximum of $\pi(p^*, a^*, q^*, s = 0)$. Instead, the optimization problem of a subscription-based platform is:

$$\max_s \pi(p, a = 0, q, s),$$

with solution s^* and maximum of $\pi(p, a = 0, q, s^*)$.

Relative profitability. We start by decomposing the difference in profits between a subscription-based business model and an ad-based business model into four components. For any scalars $\bar{a}$, $\bar{q}$, and $\bar{p}$:^{6,7}

$$\begin{aligned} \pi(p, a = 0, q, s^*) - \pi(p^*, a^*, q^*, s = 0) &= \underbrace{\pi(p^*, a^*, \bar{q}, s = 0) - \pi(p^*, a^*, q^*, s = 0)}_{\text{1. Reduced ad match quality}} \\ &+ \underbrace{\pi(\bar{p}, a^*, \bar{q}, s = 0) - \pi(p^*, a^*, \bar{q}, s = 0)}_{\text{2. No ad price differentiation}} \\ &+ \underbrace{\pi(\bar{p}, \bar{a}, \bar{q}, s = 0) - \pi(\bar{p}, a^*, \bar{q}, s = 0)}_{\text{3. No ad load discrimination}} \\ &+ \underbrace{\pi(\bar{p}, a = 0, \bar{q}, s^*) - \pi(\bar{p}, \bar{a}, \bar{q}, s = 0)}_{\text{4. Business model change}} \end{aligned} \quad (1)$$

⁶Note that: $\pi(p, a = 0, q, s^*) = \pi(\bar{p}, a = 0, \bar{q}, s^*)$ since changes in ad prices and ad match quality do not change profits when there are no ads.

⁷Note that this decomposition can be used for other outcomes beyond profits, such as engagement or consumer surplus.

The first two components correspond to the change in profits due to the elimination of microtargeting, all else equal. The first component speaks to the user side of the market: it is the difference in profits driven by the change in engagement when users face ads with a worse ad match quality. The second component speaks to the advertiser side of the market: it gives the change in profits when the platform cannot charge different prices for advertising to different segments of the market. The third component corresponds to the change in profits when the platform can no longer set different ad loads to different users, all else equal. The fourth component informs about the change in profits due to the change in business model from an ad-based to a subscription-based model.

Two objects are needed to calculate this decomposition: a counterfactual estimate of engagement, $t^\theta(a^\theta, q^\theta)$, and a counterfactual estimate of consumer surplus, $v^\theta(a^\theta, q^\theta, x)$. The goal of our experiment is to vary the ad load a^θ and ad microtargeting q^θ that individuals face to construct these counterfactuals.⁸

Optimal ad policy. We now consider the optimal ad load and optimal ad match quality that an ad-based platform should set. Rearranging the first-order condition with respect to a^θ , we obtain the optimal advertising load for users of type θ :

$$a^\theta = \frac{p^\theta t^\theta + \sum_{\tilde{\theta}} \frac{p^{\tilde{\theta}} a^{\tilde{\theta}} t^{\tilde{\theta}}}{x^{\tilde{\theta}}} \frac{\partial x^{\tilde{\theta}}}{\partial a^\theta}}{\left| \frac{\partial t^\theta}{\partial a^\theta} \right| p^\theta + \left| \frac{\partial p^\theta}{\partial a^\theta} \right| t^\theta} \quad ? \neq 0. \checkmark \quad (2)$$

Equation (2) highlights two key forces behind optimal ad allocation:

1. **Inverse elasticity.** Users whose engagement is less sensitive to ads—i.e., those with small $|\partial t^\theta / \partial a^\theta|$ —receive higher ad loads. This condition mirrors the classic inverse-elasticity pricing logic (Pigou 1920).
2. **Network spillovers.** The term $\sum_{\tilde{\theta}} \frac{p^{\tilde{\theta}} a^{\tilde{\theta}} t^{\tilde{\theta}}}{x^{\tilde{\theta}}} \frac{\partial x^{\tilde{\theta}}}{\partial a^\theta}$ captures how ads on user θ affect the platform’s revenue from other users. When θ ’s participation generates strong positive spillovers (the term is large and positive), the platform sets a lower ad load to protect that participation. Conversely, when these spillovers are weak or negative, the platform is more willing to tax θ ’s attention with ads. This parallels the logic of optimal pricing in networks (Candogan et al. 2012).

⁸Armed with these counterfactuals, we can compute the platform’s optimal subscription fee s^* in our sample. Additionally, our experiments will not produce estimates of network effects or how they impact consumer surplus, but we will use estimates from the literature, particularly from Bursztyn et al. (2025).

In terms of the optimality condition with respect to q^i , our assumptions imply that the platform will reach a corner solution and maximize the ad match quality.

3 Experiment Design

3.1 Design Overview

We conduct a field experiment to estimate user engagement responses to changes in ad load and ad targeting on Facebook. Participants install a custom browser extension that passively collects engagement data and allows us to modify the ads they see without altering organic content or the platform’s ad-serving decisions (see Farronato et al. 2024 for a general-purpose implementation of this approach).

The experiment uses two treatment-control pairs rather than a single pooled control. The first pair isolates the effect of ad quantity: *Hide* removes most ads while *Hide Control* leaves the feed unchanged. The second pair isolates the effect of ad targeting: *Replace* substitutes users’ ads with ads shown to other users while *Replace Control* applies the same technical procedure but preserves targeting. Having separate control groups ensures that comparisons are not confounded by implementation artifacts of the replacement technology.

Participants—recruited through Facebook ads—complete a six-week baseline period followed by an intervention period. We pre-registered a six-week intervention window for our main analysis, but the intervention remains active for approximately nineteen weeks total to assess persistence. Figure A1 summarizes the study flow.

3.2 Recruitment and Sample Construction

We recruit participants through Facebook ads (Figure A4 offers an example) between June 19, 2024 and December 24, 2024. Eligibility is restricted to adult U.S.-based users who use English as their primary language on Facebook and who primarily access the platform using Google Chrome. The recruitment advertisements direct interested users to a landing page that describes the study and links to the browser extension. Recruiting via Facebook ads ensures that enrollees do not use ad blockers on Facebook, since they would not have seen the recruitment ad otherwise. We describe the study as academic research on social media without disclosing any information about advertising interventions, minimizing the risk of self-selection based on ad preferences.

Enrollment requires installation of the browser extension and completion of both a baseline and an endline survey. Participants receive a \$21 gift card upon completion. Our final sample includes 1,810 users who meet a minimum Facebook usage threshold during the baseline period, run a current version of the extension, and satisfy technical requirements detailed in Appendix D.

3.3 Treatments and Intervention Implementation

3.3.1 Hide Treatment

The *Hide* treatment reduces the experienced ad load to near zero by suppressing the visual display of ads after they are delivered to the browser (but before they are delivered to the user), without altering which ads are selected by the platform’s ad auction or modifying any organic content. Figure A2 in the appendix illustrates the intervention.

The corresponding control group, *Hide Control*, experiences no modifications to the user experience. Thus, between these two groups, we exogenously vary the ad load—effectively altering the implicit price of Facebook consumption.

3.3.2 Replace Treatment

The *Replace* treatment eliminates microtargeting while holding ad quantity fixed. Ads selected for a user by the platform are replaced with ads originally shown to other users, severing the link between user characteristics and displayed advertising.

Replacement ads are drawn from a pool of previously collected English-language advertisements observed by the extension *across* the user base in the past few days. Ads are replaced on a one-for-one basis, so that the number and position of ads in the feed are identical to those in the control condition, ensuring that the visual structure and scrolling experience remain unchanged. Several constraints are imposed to ensure that treatment effects are not driven by artifacts of the replacement technology.⁹ Certain ad formats—including video ads, carousel ads, and lead-generation forms—are excluded from replacement due to technical limitations (both from being replaced and being the replacing).

Figure A3 in the appendix shows an example of an ad that replaced one that Facebook

⁹User interface elements are standardized across replaced ads, including a uniform call-to-action button (“Learn more”), and interactive features such as comments and shares are disabled for all replacement ads. In general, authors of Facebook posts can opt to disable commenting, so encountering ads with commenting disabled is not unusual. Moreover, it is rare for users to attempt commenting on or sharing ads.

intended to display. The left panel of the figure demonstrates an image of the replacement ad—it is formatted in the same way as any other Facebook ad. The right panel of the figure showcases possible interactions with the replacement ad. Users can interact naturally with replacement ads—liking, clicking through to the advertiser’s website—though commenting and sharing are disabled.

The associated control group, *Replace Control*, undergoes the identical replacement procedure except that each ad is replaced with itself, so the only exogenous variation between groups is targeting quality.

3.3.3 Common Engineering Features

To ensure that comparisons across treatment arms are not confounded by differential processing loads, all advertisements in all groups are processed through identical detection and classification routines.¹⁰

Neither intervention modifies ads that are already visible on the user’s screen (i.e., in the viewport) at the start of a Facebook session. This constraint arises because initial page loads occur concurrently with extension initialization, creating a risk that ads appear briefly before being modified. Keeping ads in the viewport at page load unchanged avoids abruptly altering content a user is actively viewing.¹¹ To prevent ads from appearing mid-scroll, the extension batches incoming feed content and holds each batch until processing is complete before releasing it to the user.¹²

3.3.4 Randomization

Randomization occurs immediately after completion of the baseline period and remains fixed for the duration of participation. Users are assigned to one of four experimental arms: *Hide*, *Hide Control*, *Replace*, or *Replace Control*. To ensure adequate sample size in the *Replace* arm despite an extension error affecting some participants (see Appendix C.2), we oversampled this condition during recruitment. Our empirical strategy accounts for the resulting unequal

¹⁰For example, *Replace* goes through the routine of selecting a random ad (which includes verifying that the image of the ad is accessible), which increases latency. To ensure identical latency between treatment and control, we force the same action in *Replace Control*, even though the ad is ultimately discarded.

¹¹In our dataset, 11% of Facebook content (posts, ads, or comments) are recorded as being in the viewport, meaning that at least a part of the element was visible on the screen at the point of rendering. The corresponding proportion for X is 15%.

¹²This happens off screen. The median time to screen for an element outside of the viewport was 4.87 seconds on Facebook, 1.72 seconds on X, and 29.7 seconds on Reddit.

allocation probabilities using inverse probability weights (see Section 3.5).¹³

Participants enter the study on a rolling basis as they respond to recruitment advertisements. For the empirical specification, we group participants into cohorts defined by their intervention start date.

The resulting sample includes 413 users in *Hide*, 414 in *Hide Control*, 538 in *Replace*, and 445 in *Replace Control*. Tables B1 and B2 in the appendix present the covariate balance between treatment and control conditions. In both cases, we assess balance on twelve covariates, including user demographics (age, sex, ethnicity, income, education, and political affiliation) and social media usage variables (number of Facebook friends and baseline engagement measures). We find no evidence of imbalance in either comparison—none of the covariates differs significantly by treatment at the 5% level and only one covariate is significant at the 10% level (in Table B2).

3.4 Data Collection and Measurement

3.4.1 Platform Activity and Ad Data

For Facebook, the extension captures the full content of the feed,¹⁴ including organic posts, advertisements, and all observable user interactions. Ads are identified using platform-specific markers and structural features, allowing us to distinguish advertising content from organic content with high accuracy.

The extension also collects feed content and ad exposure data from Reddit and X. The structural features used to identify ads differ by platform, but the procedures are analogous.

3.4.2 Ad Load Measures

We construct three measures of advertising intensity. The primary measure, *experienced ad load*, is defined as the number of advertisements *observed* by the user divided by the total volume of content (ads plus organic posts and comments) observed that day. This measure accounts for differences in consumers' typical usage patterns. For instance, it captures that consumers who spend more time in comment sections encounter fewer ads.

The second measure, *supplied ad load*, is defined analogously but uses content *served* by the platform before any intervention is applied, allowing us to test whether the platform's

¹³Results are robust to dropping users impacted by the error—see Appendix C.2.

¹⁴By “feed”, here, we mean the central portion of the screen, which excludes any advertisement that appears on the side of the window in a static fashion.

algorithm responds endogenously to our interventions.

The third measure, *feed ad load*, restricts the denominator to content appearing in the main news feed, excluding comment sections and other non-feed areas. Unlike experienced ad load, this measures what the platform shows in the feed, abstracting from user behavior. For X and Reddit, feed content is directly identifiable. For Facebook, we classify an element as feed content unless it appears in a sequence of four or more consecutive comments.¹⁵

3.4.3 Ad Match Quality

To measure how well ads match individual users, we construct an ad distance metric that captures the demographic gap between a user and the typical audience an advertiser reaches. The key idea is that a well-targeted ad should be shown to users whose characteristics align closely with the advertiser’s revealed targeting profile; a poorly targeted ad will exhibit a large discrepancy.

Specifically, let $\mathcal{K}$ denote the set of five binary demographic variables: young (age at or below the median), male, white, bachelor’s degree or higher, and household income above \$50,000. For each advertiser a and trait $k \in \mathcal{K}$, we define $\bar{X}_{ak}$ as the share of that advertiser’s impressions delivered to users with trait k , computed using baseline-period data only (Period ≤ 0) to avoid contamination by the intervention. The distance for a given user-ad pair is then

$$d_{ia} = \sqrt{\sum_{k \in \mathcal{K}} (X_{ik} - \bar{X}_{ak})^2},$$

where $X_{ik} \in \{0, 1\}$ is user i ’s value for demographic trait k . A small value of d_{ia} indicates that user i ’s demographic profile closely matches the advertiser’s typical audience, while a large value signals a poor match. We aggregate to the user-day level by averaging d_{ia} across all ads shown to user i on a given day, and normalize the resulting series to z-scores using the baseline-period mean and standard deviation to facilitate comparison across treatment conditions.

3.4.4 Engagement and Time Measures

We measure engagement along two dimensions: time spent on platforms and content consumed. For Facebook, X, and Reddit, we construct an active time measure following

¹⁵Comments in the feed do not occur in groups larger than three. A limitation is that posts with 1–3 comments are misclassified as feed elements, slightly overestimating feed content.

Beknazar-Yuzbashev et al. (2025). Active time captures periods during which the user is actively interacting with the platform—scrolling, clicking, or otherwise engaging with the feed—while excluding idle time when the browser tab remains open but the user has stepped away. To avoid truncating sessions prematurely, we append a three-minute window following the last observed interaction, accounting for brief pauses (e.g., reading a long post) that plausibly reflect continued attention.

For other websites, we employ a coarser measure: the total daily duration the browser was open on a given domain, regardless of whether the user was actively interacting. Approximately every minute, the extension records the domain visible in the address bar. Though less precise than active time, this measure allows us to track potential substitution toward competing platforms—including other social media and related websites listed in Beknazar-Yuzbashev et al. (2025) and each participant’s ten most-visited domains during baseline.

Beyond time, we record daily counts of content consumption at two levels: organic content (posts and comments viewed) and advertising content (impressions and, as a deeper measure of engagement, clicks). All measures are based on the feed data described in Section 3.4.1.

We combine time and content engagement measures into two summary indices for Facebook. The platform engagement index aggregates active time and organic content consumed, and the ad engagement index aggregates ad impressions and ad clicks. We follow the index construction method of Kling et al. (2007).

3.4.5 Long-Run Outcomes

Treatment effects may take time to fully emerge—or fade—so we extend observation beyond the pre-registered six-week intervention window. To ensure comparability across cohorts, we truncate all observation windows at a common endpoint: 176 days from enrollment. As a result, we track outcomes for approximately 19 weeks (134 days) post-randomization.

3.4.6 Survey Measures

The baseline survey collects demographic characteristics—including age, sex, race, education, household income, and political affiliation—as well as self-reported Facebook usage patterns. Both the baseline and endline survey elicit willingness to accept (WTA) payment to deactivate Facebook for four weeks.

3.5 Empirical Strategy

Our main specification exploits within-subject variation using a difference-in-differences design. For engagement outcomes recorded daily by the extension, we compare the six-week baseline period (treatments disabled) and the six-week intervention period (treatments enabled), with the individual-day as the unit of observation. Specifically, we estimate the following two-way fixed effects (TWFE) regression:

$$Y_{i,t} = \alpha_i + \delta_{t,C_i} + \beta D_{i,t} + \text{Errors}_{i,t} + \varepsilon_{i,t}, \quad (3)$$

where α_i are individual fixed effects, δ_{t,C_i} are cohort $\times$ period fixed effects, and $D_{i,t}$ equals one if user i is assigned to treatment and period t falls in the intervention window. We index time in days relative to each individual’s intervention start date, with $t = 0$ denoting the final day of the baseline period. The coefficient β identifies the treatment effect. We include cohort $\times$ period fixed effects, as opposed to just period fixed effects, to ensure that our estimation is robust to the staggered roll-out of the treatment—see Cengiz et al. (2019) for an application of this approach. We include controls for browser extension errors described in Appendix C and their interactions with treatment.¹⁶

We estimate two main comparisons: *Hide* versus *Hide Control* and *Replace* versus *Replace Control*. All regressions use inverse probability weights to account for the oversampling of the Replace condition described in Section 3.3.4. To increase statistical power, we use Driscoll and Kraay (1998) standard errors—they account for both serial and cross-sectional dependence by exploiting the long panel in our study (see Cameron and Miller 2015; Hoechle 2007 provides evidence on finite-sample performance).

For outcomes measured only in surveys (baseline and endline), we estimate a two-period difference-in-differences: the outcome regressed on a treatment indicator, an endline indica-

¹⁶We account for three types of possible errors: (i) website update—Facebook made a sudden change to the HTML code of the website, which resulted in issues such as advertiser names displaying incorrectly, (ii) faulty replacement—eligible ads in the *Replace* group were incorrectly replaced, (iii) faulty hiding—the hiding intervention was incorrectly enabled during parts of the baseline period for a random subset of participants. Such errors are common in browser-extension-based research, as extensions must be adapted to ongoing changes in platform HTML structure. The share of user-day observations in which participants experienced at least one instance of each error was small: 0.049 for the website update, 0.005 for faulty replacement, and 0.004 for faulty hiding for the pre-registered study period. Depending on whether the error affected all treatment groups, occurred for different individuals on different days, or occurred during both the baseline and intervention periods, some variables in $\text{Errors}_{i,t}$ may be absorbed by the intervention start $\times$ period fixed effects. See Appendix C for a more comprehensive description of the errors and related robustness analysis.

tor, and their interaction, with individual fixed effects.

To examine heterogeneity by baseline characteristics, we augment the main specification:

$$Y_{i,t} = \alpha_i + \delta_{t,C_i,M_i} + \beta D_{i,t} + \phi(D_{i,t}M_i) + \text{Errors}_{i,t} + \varepsilon_{i,t}. \quad (4)$$

where M_i indicates whether user i is above the median of the characteristic of interest. The coefficient β captures treatment effects for below-median users; $\beta + \phi$ captures effects for above-median users. Cohort $\times$ period fixed effects are interacted with M_i to allow differential trends.

4 Results

4.1 Descriptive Evidence on Ad Loads

The model in Section 2 highlights three channels that may sustain advertising model dominance: user insensitivity to ad loads, improved match quality through microtargeting, and ad load personalization across users. We begin by documenting descriptive patterns in ad loads that speak directly to the third channel.

4.1.1 Ad Load Heterogeneity

We first characterize ad loads on three major social media platforms. Figure 1 presents distributions of feed ad loads consumers see on average in the baseline period. The average consumer faces a 12.8% ad load on Facebook, 11.4% ad load on X, and 8.7% ad load on Reddit. Ad loads vary substantially across users on all three platforms: the standard deviations of ad loads across users are 4.6, 6.8, and 3.7 percentage points on Facebook, X, and Reddit, respectively.

Most of this cross-user variation reflects persistent individual differences rather than day-to-day fluctuations. Decomposing the variance in feed ad loads into individual and day components (Figure A6), we find that individual fixed effects explain 67% of the variance on Facebook, 70% on X, and 72% on Reddit. This dominance of user-specific differences suggests systematic ad-load personalization.

We find a consistent pattern using experienced ad loads, a measure that accounts for typical user activity on the platform. Figure A5 presents experienced ad loads both across user-day pairs (A5a) and across users (A5b); the two distributions are similar, confirming

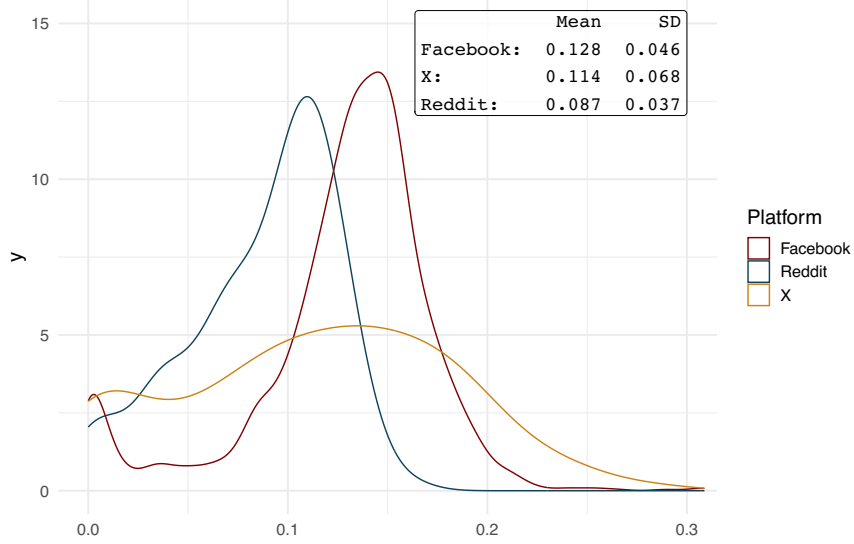


FIGURE 1: DISTRIBUTION OF INDIVIDUAL FEED AD LOAD BY PLATFORM

Note: The figure presents the distributions of the feed advertising load across users on Facebook, Reddit and X, averaged at the user level over the entire untreated period. The feed ad load is defined as the number of ads as a share of total content in the feed (both posts and comments) shown to the user on the platform. The browser extension can directly identify feed elements on Reddit on X. On Facebook, we consider an element to be in the feed unless it is a part of a sequence of at least four consecutive comments—comments in the feed do not occur in groups larger than three. The figure relies on the subsample of users who viewed at least 50 posts and comments throughout the relevant period.

that the heterogeneity documented above is driven by across-user rather than within-user variation.

4.1.2 Ad Load Correlates

Which user characteristics predict ad loads? Figure 2 reports general dominance—the average incremental R^2 each covariate contributes across all possible subset models—for a set of demographic and technology-use covariates predicting feed ad loads. Age is the strongest predictor by a wide margin: age and age squared rank first and second. The importance of age is consistent with older users being both less price-elastic and more valuable to advertisers (Gentzkow et al. 2024). Facebook desktop usage share, income, political affiliation, and platform valuation are weaker predictors.

The observed patterns are broadly consistent with our model’s predictions (Section 2) on optimal ad-load setting. We pre-registered three tests, shown in Figure 3. First, the model predicts that users who generate larger network spillovers receive lower ad loads, as the platform protects their participation; consistent with this, users with above-median

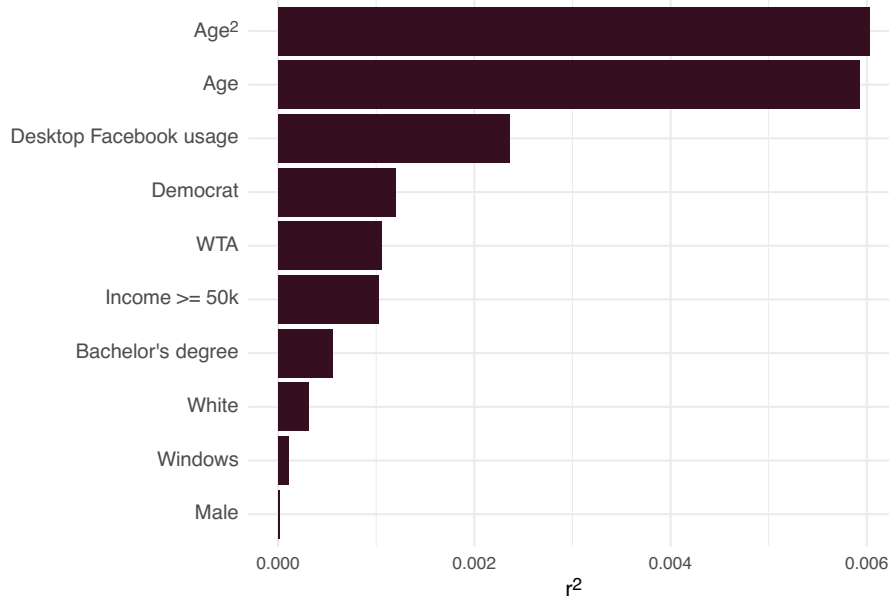


FIGURE 2: GENERAL DOMINANCE OF FEED AD LOAD PREDICTORS ON FACEBOOK

Note: The figure reports general dominance for a broad set of covariates predicting feed ad loads on Facebook, i.e., the average incremental R^2 a variable contributes when considering all possible subset models. The included variables are as follows: (1) age in years, (2) age squared, (3) an indicator for having at least a bachelor’s degree, (4) an indicator for being a Democrat, (5) desktop share of Facebook usage (from the baseline survey), (6) an indicator for having household income above \$50,000, (7) an indicator for being male, (8) an indicator for identifying as white/Caucasian, (9) an indicator for using Windows operating system, (10) willingness to accept to deactivate Facebook for four weeks during the baseline survey (WTA).

Facebook friends face significantly lower ad loads. Second, the model predicts that more ad-sensitive users receive lower ad loads; users with above-median predicted elasticity—estimated via generalized random forests (Athey et al. 2019) using the differences in *Hide* and *Hide Control* groups—receive feed ad loads nearly 2 percentage points lower, a larger gap than in the network test. Third, the model is ambiguous about ad value, since higher-value users generate more revenue per impression but may also be more worth retaining. Empirically, high-value users receive higher ad loads, suggesting that margin considerations dominate retention motives.

4.1.3 Ad Loads Across Platforms

Do different platforms charge the same users similar “attention prices”? Individual ad loads correlate positively across Facebook, X, and Reddit—most strongly between X and Reddit ($r = 0.45$, $p = 0.003$), more modestly between Facebook and X ($r = 0.15$, $p = 0.03$) and between Facebook and Reddit ($r = 0.12$, $p = 0.13$). This suggests that platforms tailor ad

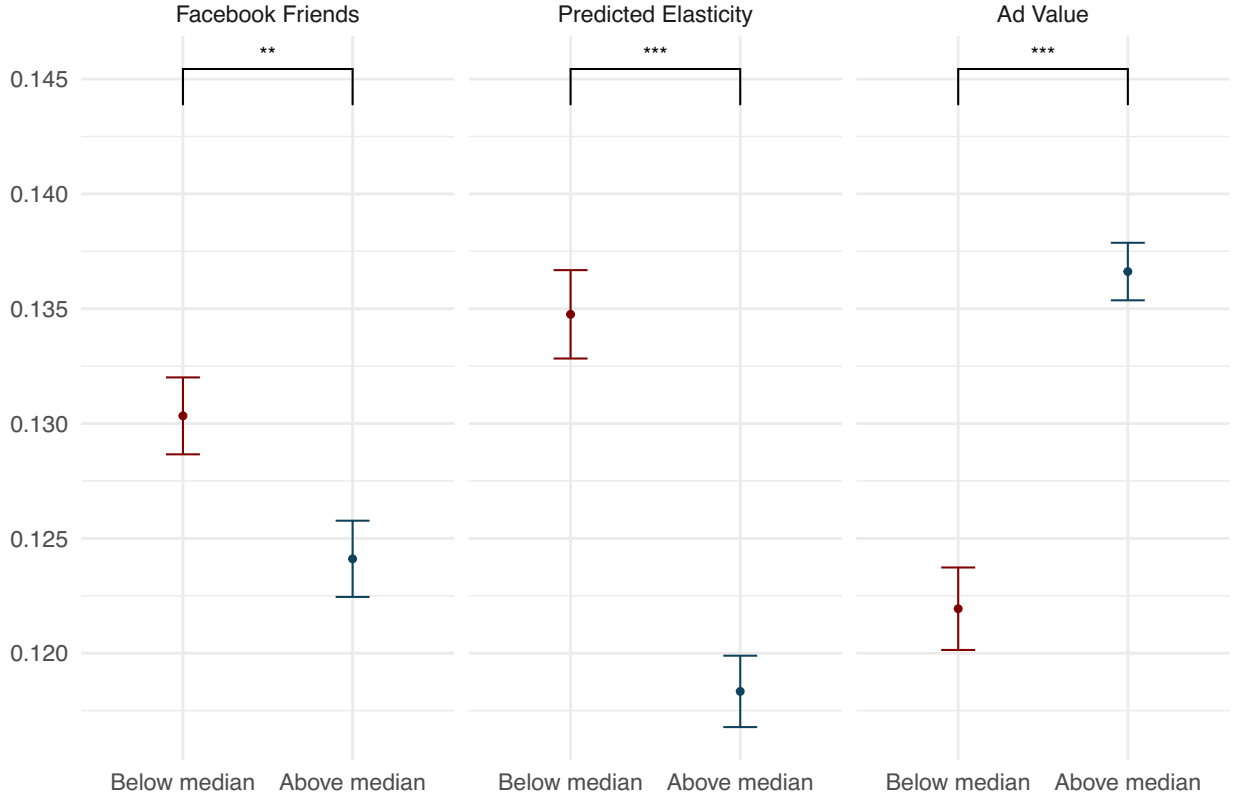


FIGURE 3: TESTING PREDICTIONS ON AD LOADS HETEROGENEITY

Note: The figure reports estimates from regressions of individual-level average feed ad load on an indicator equal to one if the heterogeneity variable exceeds its median. The heterogeneity variables are: (1) ad value, (2) number of Facebook friends, and (3) predicted ad load elasticity. Ad value measures how valuable it is for the platform to sell ad slots targeting a given user, based on demographics. To construct individual-level ad value, we use \$100 advertising campaigns targeted by sex, age, and high/low-income ZIP codes and obtain Facebook’s estimates of cost per click; ad value is then computed as the product of cost per click and the individual’s ad click rate. Predicted ad load elasticities are computed using a generalized random forest approach, which estimates conditional average treatment effects of the hiding intervention as functions of individual covariates. The figure reports point estimates with 95% confidence intervals.

loads to similar user characteristics.

Platforms also show some overlap in which ad topics they show to the same users. We classify advertisers into 26 categories based on the IAB Content Taxonomy and compute, for each pair of platforms, within-topic correlations in how users are targeted.¹⁷ The resulting correlations (Figure A10) are positive and significant for all platform pairs.¹⁸ They are also modest: 0.22 for Facebook–Reddit, 0.14 for X–Reddit, and 0.10 for Facebook–X. Platforms

¹⁷Specifically, we classify each advertiser using Tier1 of the IAB Content Taxonomy v1.0. For each advertiser, we send its name and up to five recent ad texts to GPT-5-mini and repeat the classification three times, taking the majority vote as the final category. See Figure A7 in the appendix for distributions.

¹⁸To obtain platform-level measures, for each pair of platforms we take the mean of the diagonal correlation coefficients, which capture within-topic correlations in how users are targeted across platforms.

thus share some commonality in how they target users by topic, but also appear to rely on distinct signals.

4.2 Experimental Results

4.2.1 First Stage

Hiding Intervention and Ad Load. Figure 4 shows the average Facebook ad load experienced by treatment condition over the study period. At baseline, ad load averaged 0.128 with no significant differences across groups. Following the intervention, ad load in the *Hide* condition dropped sharply to 0.017—an 87% reduction—and remained stable throughout. The *Hide Control* condition held steady at 0.137, confirming that the intervention affected only treated users. The difference-in-differences estimate indicates a reduction in experienced ad load of 11.3 pp (82.5%, or 1 SD) relative to *Hide Control* (Table B3).

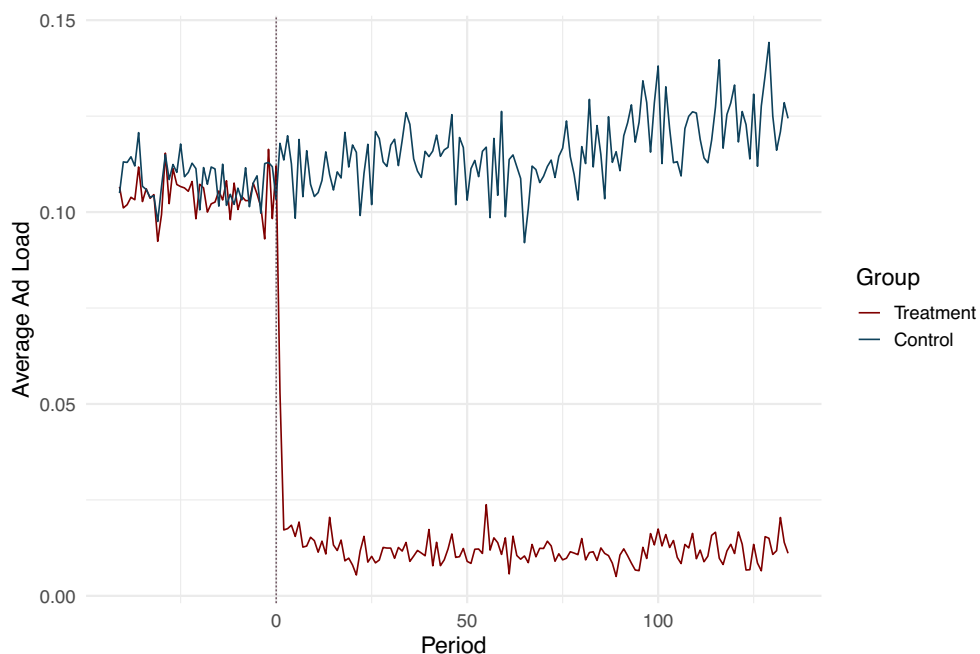


FIGURE 4: AVERAGE ADVERTISING LOADS ON FACEBOOK

Note: The figure shows the average daily advertising load on Facebook for the *Hide* condition and the *Hide Control* condition. Advertising load is defined as the number of ads shown as a share of all Facebook posts and comments viewed by the user. The x-axis represents study periods, measured in days relative to each individual’s intervention start date. The vertical dashed line marks the final day of the pre-intervention period (period 0).

The platform’s ad-serving algorithm did not respond to the intervention. Supplied ad load—measured before our filtering—was virtually unchanged between *Hide* and *Hide Con-*

trol (0.3 pp, Table B3, column 3, panel A), and long-term estimates are nearly identical (Table B4). Facebook continued to serve the same share of ads to treated users despite their dramatically different experienced ad loads.

Replacement Intervention and Microtargeting. Figure 5 shows average ad distance—our measure of ad-user match quality—over time. During the baseline period, ad distance evolved similarly across treatment groups. Immediately after the intervention, ad distance in the *Replace* condition jumped and remained elevated, confirming that the replacement intervention successfully degraded microtargeting. The difference-in-differences estimate indicates an 11% increase in average ad distance (Table B3, column 2, panel B).

Unlike the hiding intervention, replacement triggered a mild algorithmic response: Facebook adjusted the ads it supplied, with the supplied ad distance by 1.5% higher (Table B3, column 4, panel B). This pattern persists in the long-term sample (Table B4). The asymmetry likely reflects how the algorithm weights recent interactions—as users engaged with different ad content, the platform updated its model of their preferences. The pattern suggests that Facebook treats users’ ad-load elasticities as stable but allows inferred ad preferences to evolve over time.

4.2.2 Effects of Hiding Ads

Ad Engagement. Figure 6(a) presents the short-run treatment effects; Table B5 reports full regression results. The hiding intervention sharply reduced ad engagement: impressions fell by 95% (0.28 SD) and clicks by 58% (0.11 SD). The combined ad engagement index declined by 0.19 SD. Long-run effects over 19 weeks (Figure 6(b); Table B6) are consistent in both direction and magnitude.

Platform Engagement. While ad engagement fell, overall platform engagement increased. Using our pre-registered specification in levels, the short-term effects are modest: content consumption increased by 5.6 pieces per day (9%, or 0.03 SD; $p < 0.1$) and active time by 0.65 minutes per day, though the latter is not statistically significant (Figure 6(a); Table B5). Long-term estimates are similar in magnitude: content consumption increased by 0.04 SD ($p < 0.01$) and active time by 0.57 minutes per day (Figure 6(b); Table B6). The overall platform engagement index rose by 0.03 SD in the short run and 0.02 SD in the long run ($p < 0.1$). The effects emerge early and persist over time.

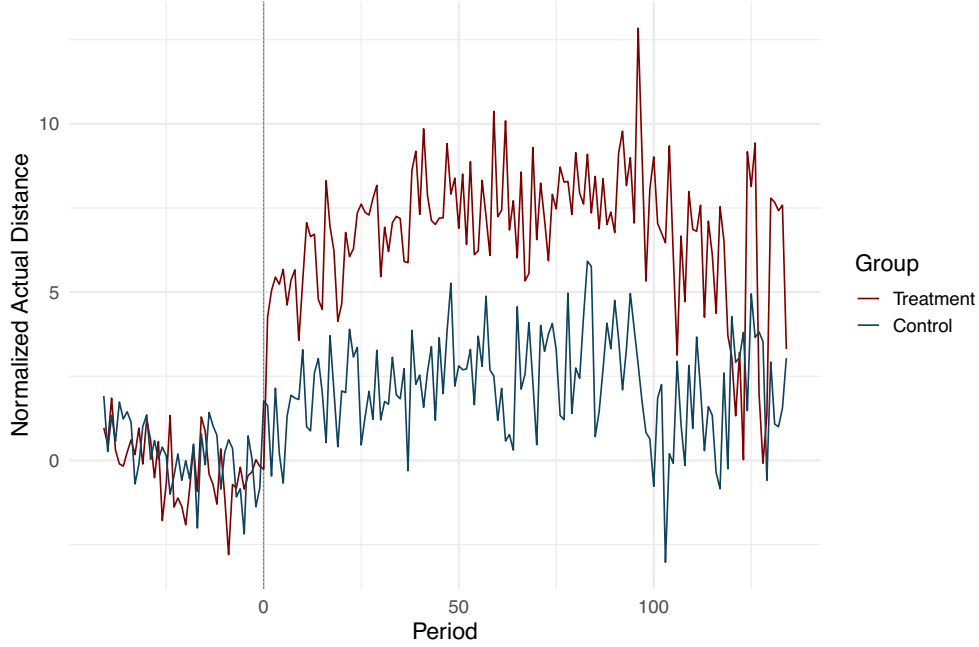


FIGURE 5: NORMALIZED AD DISTANCE ON FACEBOOK

Note: The figure displays the average daily ad distance on Facebook, disaggregated by treatment condition (Replace vs. Replace Control). Ad distance is defined as the Euclidean distance between a user’s characteristics and the average demographic profile targeted by that advertiser, based on baseline data. The values on the y-axis are normalized using the mean and standard deviation from the baseline period. The x-axis represents study periods, measured in days relative to each individual’s intervention start date. The vertical dashed line marks the final day of the pre-intervention period (period 0).

Outliers and skewness in the levels specification reduce precision of our estimates (see Section 4.2.6). We address this with two alternative specifications: an inverse hyperbolic sine (*asinh*) transformation and a levels specification trimming the top percentile. Both yield significant positive effects on active time and content consumption in both the short and long run (Tables B7, B9, B8, and B10). Under these specifications, the platform engagement index rises by 0.05-0.07 SD ($p < 0.01$)—roughly double the baseline estimate—and active time increasing by roughly 1.6 minutes per day in the long run.

Elasticities. We translate these effects into an elasticity: the ratio of the percentage change in active time to the percentage change in ad load. Using our difference-in-differences estimates (Tables B3 and B5), the short-run elasticity is:

$$\text{Elasticity} = \frac{\% \Delta \text{Active Time}}{\% \Delta \text{Ad Load}} = -0.07.$$

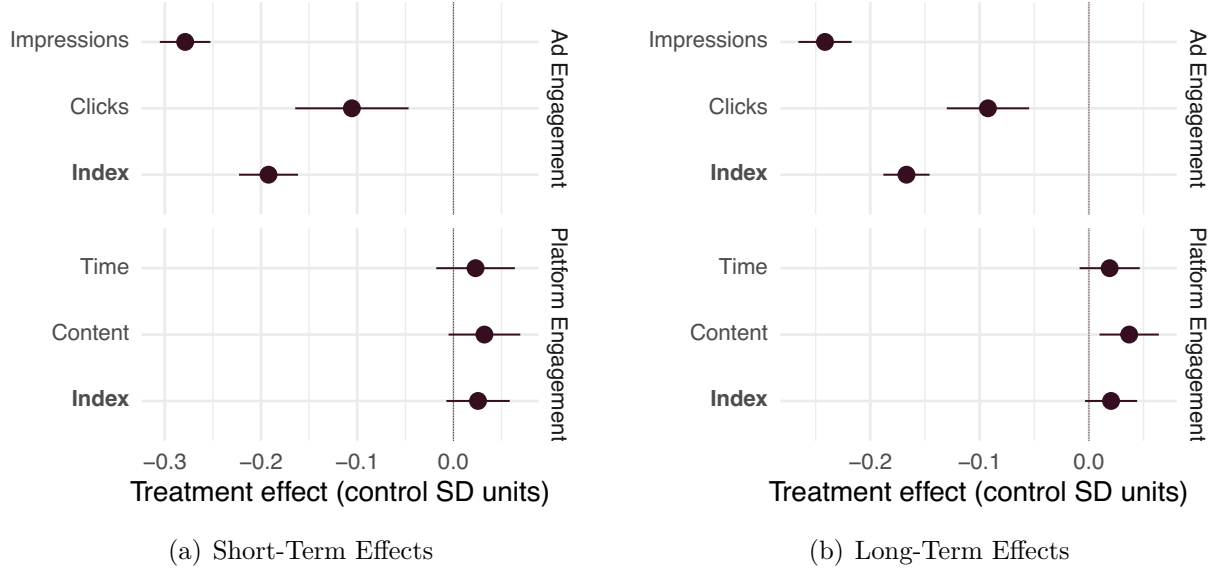


FIGURE 6: OWN-PLATFORM ENGAGEMENT EFFECTS OF HIDING INTERVENTION

Note: The figure displays estimated treatment effects of the hiding intervention on Facebook using Equation 3 and our main experimental sample. The dependent variables are: (i) number of ad impressions, (ii) number of ad clicks, (iii) an ad engagement index based on (i) and (ii), (iv) active time spent on the platform, (v) number of posts and comments shown to the user, and (vi) a platform engagement index based on (iv) and (v). **Panel A** shows results based on the six-week intervention period, whereas **Panel B** shows results based on a longer intervention period (134 day-periods or 19.1 weeks). The unit of observation is the individual-day, measured relative to the intervention date. We include 95% confidence intervals based on Driscoll-Kraay standard errors. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day.

The *asinh* specification yields -0.16 (Table B7).¹⁹ Both estimates are below 0.2 in absolute value, indicating highly inelastic demand. This accords with Brynjolfsson et al. (2025), who reach a similar conclusion.

4.2.3 Effects of Replacing Ads

Ad Engagement. Figure 7(a) presents short-term effects of the replacement intervention; Table B5 reports full results. Replacement reduced ad engagement: impressions fell by 0.05 SD and clicks by 0.25 SD. The combined index declined by 0.15 SD. Long-term effects are similar, with the index falling by 0.18 SD (Figure 7(b); Table B6). These results indicate

¹⁹To express treatment effects estimated on *asinh*-transformed outcomes as percentage changes in the original (level) units, we use the following formula:

$$\% \Delta Y = \frac{\sinh(m + \beta) - \sinh(m)}{\sinh(m)} \times 100,$$

where β is the estimated treatment effect and m is the mean of the *asinh*-transformed outcome variable. This expression maps the estimated effect back to the original scale and expresses the change as a percentage relative to the mean.

that microtargeting is a key driver of ad engagement.

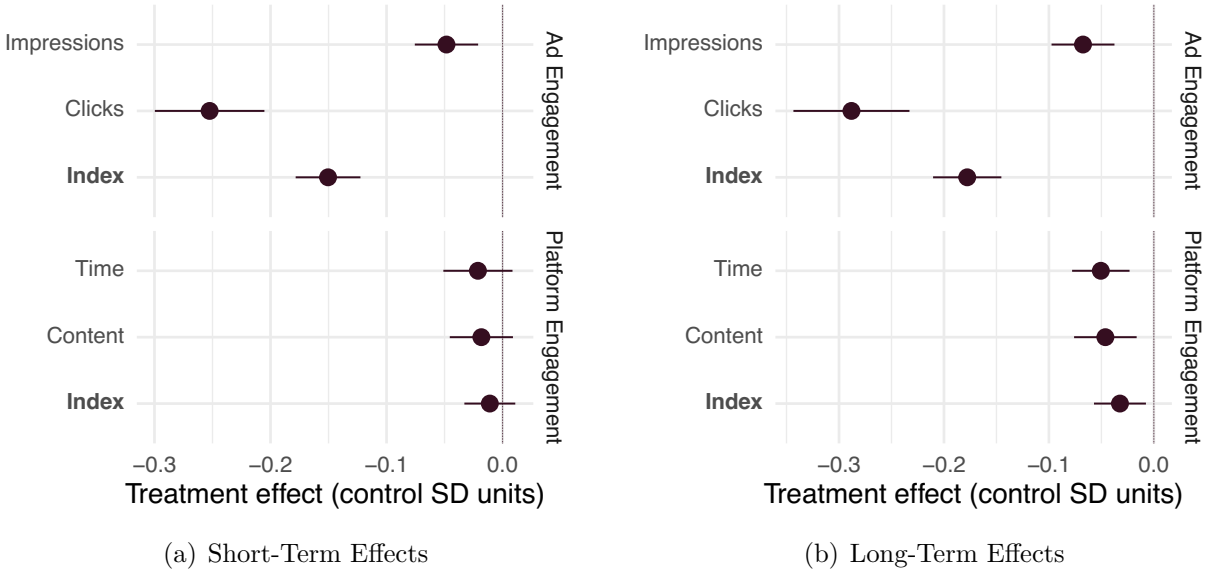


FIGURE 7: OWN-PLATFORM ENGAGEMENT EFFECTS OF REPLACEMENT INTERVENTION

Note: The figure displays estimated treatment effects of the replacement intervention on Facebook relative to *Replace Control* using Equation 3 and our main experimental sample. The dependent variables are: (i) number of ad impressions, (ii) number of ad clicks, (iii) an ad engagement index based on (i) and (ii), (iv) active time spent on the platform, (v) number of posts and comments shown to the user, and (vi) a platform engagement index based on (iv) and (v). **Panel A** shows results based on the six-week intervention period, whereas **Panel B** shows results based on a longer intervention period (134 day-periods or 19.1 weeks). The unit of observation is the individual-day, measured relative to the intervention date. We include 95% confidence intervals based on Driscoll-Kraay standard errors. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day.

Platform Engagement. In the short run, the replacement intervention had no statistically significant effect on platform engagement, a pattern that holds across specifications (Figure 7(a); Table B5). Over the longer run, however, clear negative effects emerged (Figure 7(b) and Table B6). Active time fell by 1.7 minutes per day (13%, 0.05 SD, $p < 0.01$) and content consumption by 9.4 posts per day (13%, 0.05 SD, $p < 0.01$), with the platform engagement index declining by 0.03 SD ($p < 0.05$). These long-run results are robust to trimming (0.04 SD, $p < 0.01$) and *asinh* specifications (0.06 SD, $p < 0.01$).

The delayed emergence of platform engagement effects suggests that users update their beliefs about the platform experience gradually as they encounter less relevant ads—an important consideration underscoring the need for long-duration interventions to inform policy. The replacement intervention thus reduced both ad and platform engagement, implying that both consumers and the platform may benefit from ad microtargeting.

4.2.4 Substitution Effects

Figure 8(a) presents effects of the hiding intervention on time spent on competing platforms. We find little evidence of substitution: the intervention does not significantly affect time on X and Reddit, YouTube, or the combined set of pre-registered social media platforms. However, time on users’ top 10 most-visited domains increased—possibly a “linking effect,” whereby greater Facebook engagement drives traffic to sites frequently shared in users’ feeds.

The replacement intervention yields no clear substitution pattern either (Figure 8(b)). Active time on X and Reddit increased marginally, as did time on top-visited domains, but time on YouTube fell and effects on the broader set of social media platforms are insignificant.

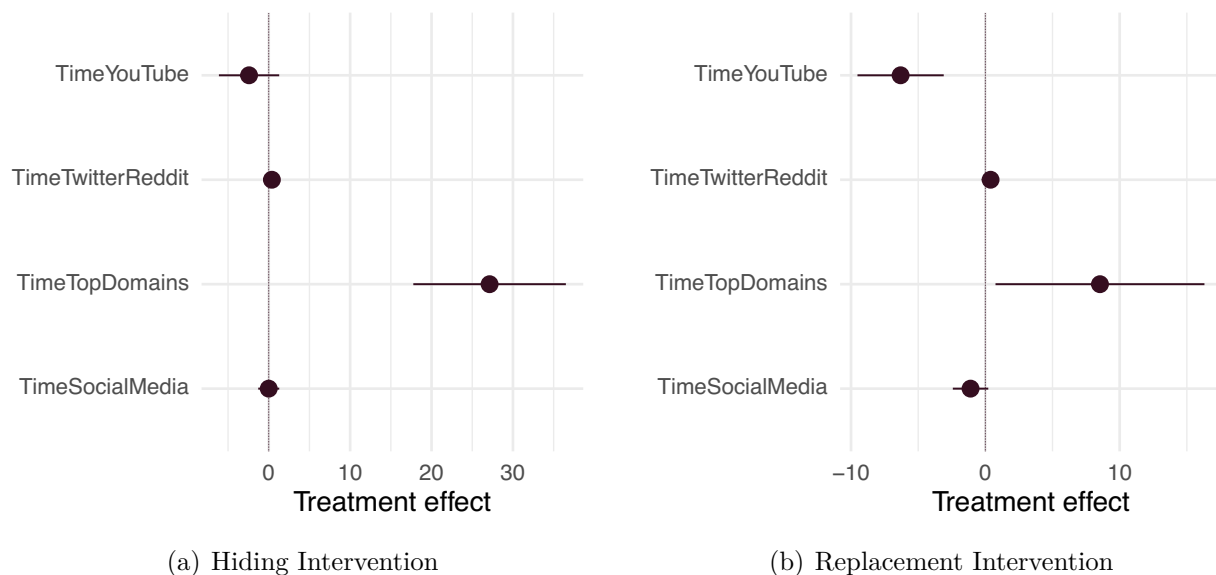


FIGURE 8: IMPACTS ON ENGAGEMENT WITH NON-TREATED PLATFORMS

Note: **Panel A** reports estimated treatment effects of the hiding intervention on Facebook, relative to the *Hide Control* group, based on Equation 3 and the main experimental sample. **Panel B** presents treatment effects of the replacement intervention, relative to the *Replace Control* group. The dependent variables are: (i) time spent on YouTube; (ii) combined active time on X and Reddit; (iii) time spent on the top 10 domains as determined from baseline data; and (iv) time spent on a set of non-treated social media platforms defined in Beknazar-Yuzbashev et al. (2025). All outcome variables are inverse hyperbolic sine (asinh) transformed. The unit of observation is the individual-day, indexed relative to the intervention start date. Regressions include 95% confidence intervals based on Driscoll-Kraay standard errors. All estimates are weighted using inverse probability weights reflecting group assignment probabilities conditional on recruitment day.

4.2.5 Platform Valuation

Figure 9 presents effects on platform valuation, measured as willingness to accept (WTA) payment to deactivate Facebook for four weeks. Neither intervention significantly affects

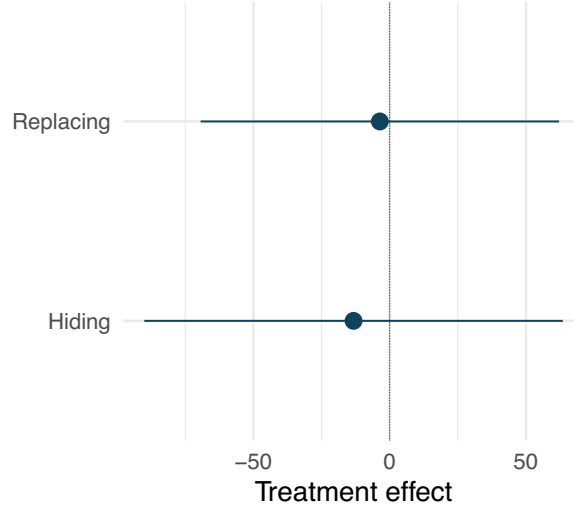


FIGURE 9: EFFECTS OF INTERVENTIONS OF PLATFORM VALUATION

Note: The figure displays the estimated treatment effects of the hiding and replacement interventions on individuals' willingness to accept (WTA) the deactivation of social media for four weeks. WTA is measured in both the baseline and endline surveys. The estimates are based on a regression of WTA on the treatment group indicator, the endline survey indicator, and their interactions, with individual fixed effects. The unit of observation is the individual, and 95% confidence intervals are provided, based on standard errors clustered at the individual level. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day.

WTA. The null result for hiding is consistent with Brynjolfsson et al. (2025).²⁰ We extend their finding by showing that degraded microtargeting also leaves platform valuation unchanged.

4.2.6 Heterogeneous Effects

Figure 10 presents treatment effects of the hiding intervention by baseline engagement (above vs. below median active time). Negative effects on ad engagement concentrate among heavy users: the ad engagement index falls by 0.35–0.40 SD for above-median users, with smaller but detectable effects for below-median users.

Platform engagement effects are less clear-cut. Confidence intervals for above-median users are wide, driven by outliers among heavy users. As before, we address this with an *asinh* specification (Figure A8). Both groups show significant positive effects on platform engagement of roughly 0.05 SD.

Figure 11 presents corresponding results for the replacement intervention. Heavy users

²⁰Our estimates of the effects of the interventions on WTA are less precise than theirs due to a smaller sample size.

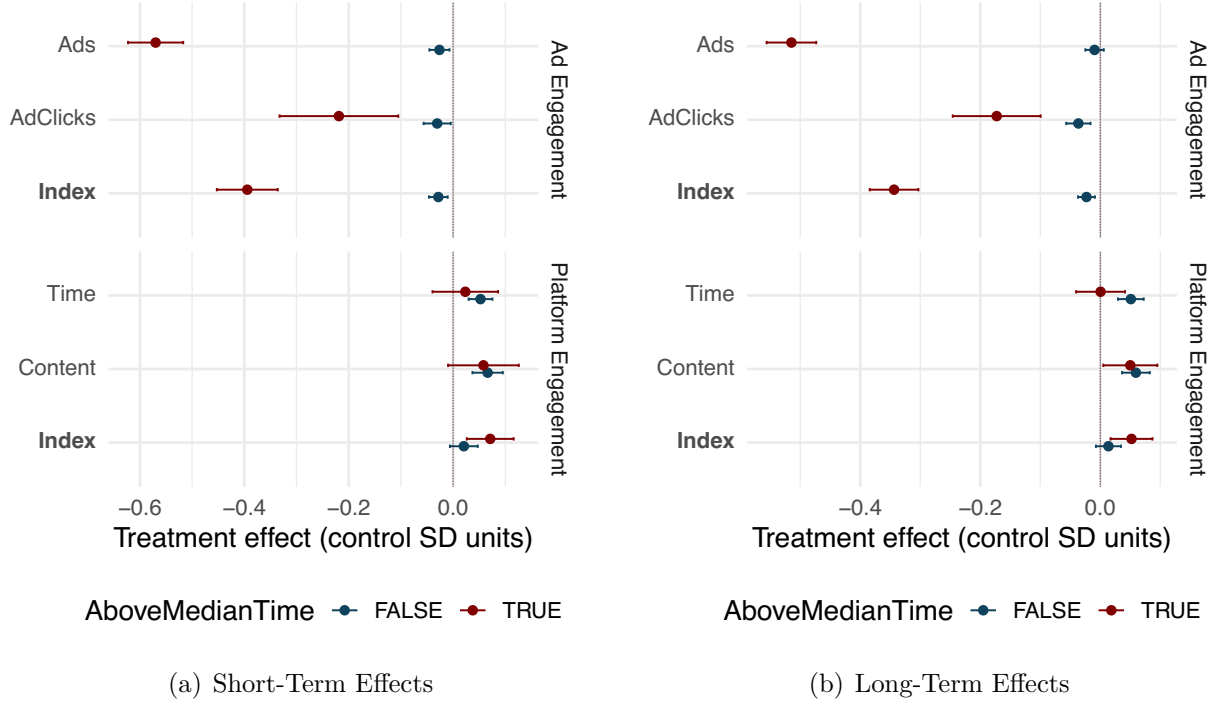


FIGURE 10: HETEROGENEITY OF ENGAGEMENT EFFECTS: HIDING

Note: The figure displays estimated treatment effects of the hiding intervention on Facebook relative to *Hide Control* separately for users whose baseline active time on Facebook was above and below the median. The estimation is based on Equation 4, with $\beta + \phi$ representing the above-median coefficient and β representing the below-median coefficient. The regression relies on the main experimental sample. The dependent variables are: (i) number of ad impressions, (ii) number of ad clicks, (iii) an ad engagement index based on (i) and (ii), (iv) active time spent on the platform, (v) number of posts and comments shown to the user, and (vi) a platform engagement index based on (iv) and (v). **Panel A** shows results based on the six-week intervention period, whereas **Panel B** shows results based on a longer intervention period (134 day-periods or 19.1 weeks). The unit of observation is the individual-day, measured relative to the intervention date. We include 95% confidence intervals based on Driscoll-Kraay standard errors. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day.

again drive the negative effects on ad engagement: ad clicks fall by more than 0.5 SD for above-median users. Short-run effects on platform engagement are insignificant for both groups, consistent with Section 4.2.3. In the long run, however, above-median users show clear declines—active time and content consumption both fall by nearly 0.1 SD. Results are similar under the asinh specification (Figure A9).

4.3 Robustness

This section discusses potential concerns and robustness checks. Section 4.3.1 shows that attrition does not vary by treatment assignment. Section 4.3.2 demonstrates robustness to alternative specifications. Appendix C confirms that results are unchanged when excluding participants who experienced browser extension errors.

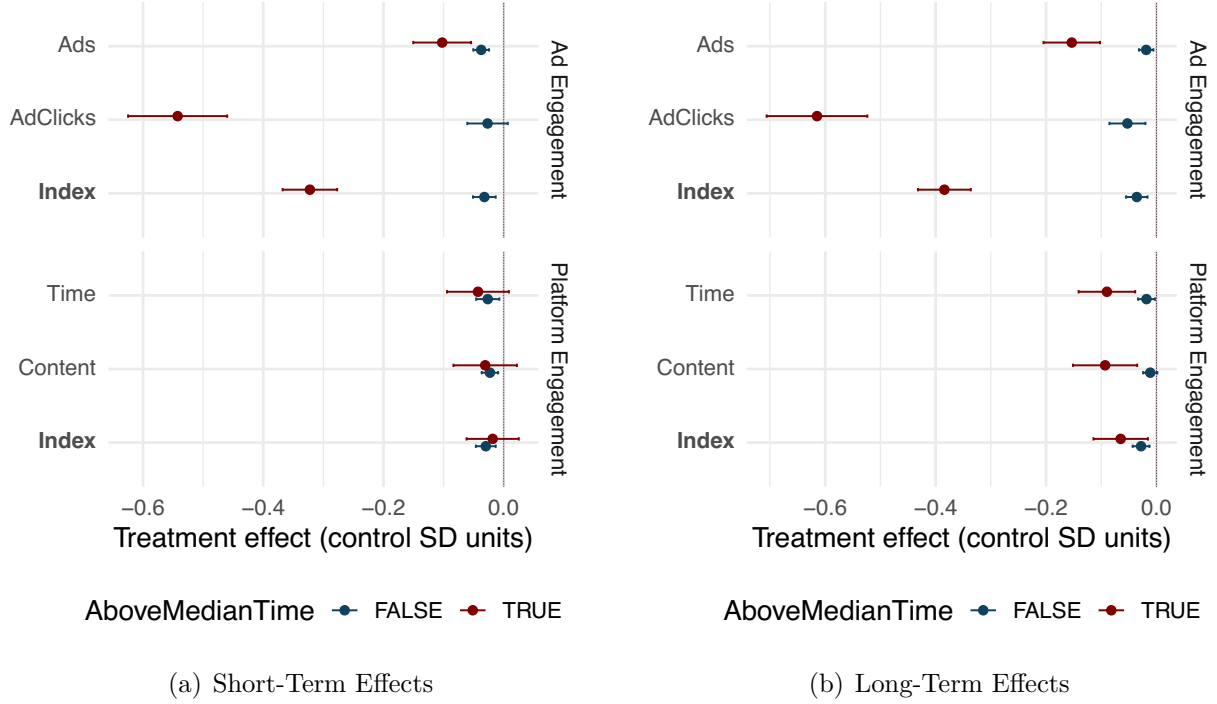


FIGURE 11: HETEROGENEITY OF ENGAGEMENT EFFECTS: REPLACEMENT

Note: The figure displays estimated treatment effects of the replacement intervention on Facebook relative to *Replace Control* separately for users whose baseline active time on Facebook was above and below the median. The estimation is based on Equation 4, with $\beta + \phi$ representing the above-median coefficient and β representing the below-median coefficient. The regression relies on the main experimental sample. The dependent variables are: (i) number of ad impressions, (ii) number of ad clicks, (iii) an ad engagement index based on (i) and (ii), (iv) active time spent on the platform, (v) number of posts and comments shown to the user, and (vi) a platform engagement index based on (iv) and (v). **Panel A** shows results based on the six-week intervention period, whereas **Panel B** shows results based on a longer intervention period (134 day-periods or 19.1 weeks). The unit of observation is the individual-day, measured relative to the intervention date. We include 95% confidence intervals based on Driscoll-Kraay standard errors. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day.

4.3.1 Attrition

Figure A14 in the appendix shows survival rates by treatment group over time. We measure attrition conservatively: an individual i is classified as having survived until period t only if they were active on day t or later. Survival rates at the end of the six-week intervention exceed 75% in all arms, ranging from 76.1% (*Replace*) to 76.9% (*Replace Control*), and the trajectories are parallel across conditions throughout the study period. Table B11 provides formal tests: regressions of survival on treatment assignment yield no significant pairwise differences, and interactions with predicted survival probability are uniformly insignificant.²¹

Because we report long-term outcomes, we also examine attrition over the full observation

²¹The dependent variable is capped at day 42, the final day of the six-week intervention; baseline periods are indexed as zero and below.

window. Figure A14 shows that survival trajectories remain parallel well beyond the six-week intervention (dashed vertical line). Table B12 confirms this formally: regressions of uncapped dropout timing on treatment indicators reveal no differential attrition. We conclude that attrition is unlikely to explain either the short- or long-run results.

4.3.2 Alternative Specifications

Skewness and Outliers. As noted in Sections 4.2.2 and 4.2.3, our main specification uses untransformed outcomes, which are right-skewed. We address this with two alternatives: an inverse hyperbolic sine transformation (Tables B7 and B8) and top-1% trimming (Tables B9 and B10). Both yield improved precision and reinforce the main findings: the hiding intervention increases platform engagement in the short and long term, while the replacement intervention reduces it in the long term.

Alternative Standard Errors. Our main specification uses Driscoll-Kraay standard errors, which account for both serial and cross-sectional dependence. As a more conservative alternative, we cluster at the individual level (Tables B13 and B14). Under clustering, ad engagement effects remain significant for both interventions (1% for hiding, 10% for replacement), though platform engagement effects lose significance in the levels specification. Under the *asinh* transformation, the positive effects of hiding on platform engagement and the long-term negative effects of replacement are robust to individual-level clustering (Tables B15 and B16).

4.4 External Validity

Our sample differs from the full Facebook population in two respects that are worth noting. The extension operates only on desktop browsers, whereas a substantial share of Facebook usage occurs on mobile devices; to the extent that desktop and mobile users differ in their sensitivity to ads, our estimates may not extrapolate to the full user base. Our sample also consists of U.S.-based, English-speaking users who responded to a Facebook recruitment ad, introducing potential self-selection. Our results are therefore most directly informative about active Facebook users who engage with platform content, including advertising. These limitations are common to browser-extension-based designs and should be weighed against the advantage of observing actual platform behavior without relying on self-reports or platform cooperation.

Independent evidence suggests that neither limitation materially affects our main conclusions. Our core finding—that engagement is highly inelastic to ad loads—is consistent with Brynjolfsson et al. (2025), who run a server-side experiment removing ads for a random 0.5% of *all* Facebook users over nine years. Their design addresses both concerns above: it captures mobile and desktop usage, and their sample is drawn randomly from the full user base rather than recruited through advertising. The agreement of elasticity estimates across two very different experimental designs suggests that our findings are not driven by the desktop or self-selection restrictions.

5 Counterfactual Monetization

We now use the experimental estimates to simulate counterfactual monetization regimes and quantify the decomposition in (1). We compare outcomes under three scenarios: the observed advertising-based model with personalized ad loads and microtargeting, a subscription-based model with uniform monetary pricing and no ads, and a hybrid “opt-in” alternative where users choose between an ad-supported experience and a paid ad-free tier. For each regime, we evaluate profits, time spent on the platform, consumer surplus, and platform demand (measured as the share of users with positive consumer surplus).

5.1 Description

The counterfactual construction proceeds in four steps. First, we estimate heterogeneous causal effects of ad load and ad targeting on user outcomes using the randomized variation induced by our experiment. For each outcome, we collapse the daily panel into user-level differences between the six-week intervention period and the six-week baseline period and estimate conditional average treatment effects (CATEs). The causal effect of ad load is estimated using an instrumental forest (Athey et al. 2019), where changes in ad load are instrumented by assignment to the *Hide* versus *Hide Control* groups. The causal effect of ad targeting is estimated using a causal forest based on assignment to the *Replace* versus *Replace Control* groups. In both cases, the feature set includes baseline demographic characteristics, baseline platform usage and ad exposure, and indicators for browser-extension errors. We estimate CATEs in-sample for each comparison group and then predict them out-of-sample for the other treatment groups. Figure A11 presents the distribution of predicted CATEs under an ad-hiding counterfactual for content viewed, time spent on the platform, and

willingness to accept (WTA) to deactivate the platform.

Second, we use the estimated user-level treatment effects to construct counterfactual outcomes via a local linear approximation around observed post-intervention outcomes. For each individual, we predict outcomes under counterfactual ad loads and ad targeting regimes. Concretely, we compute:

$$y_i(a'_{it}, q'_{it}) = \bar{y}_i + \hat{\gamma}_i^a(a'_i - \bar{a}_i) + \hat{\gamma}_i^q(q'_{it} - q_{it}),$$

where $\bar{y}_i$ and $\bar{a}_i$ denote the (average) observed outcome and ad load during the intervention period, respectively, q_{it} is a dummy variable indicating whether the user experienced a reduction in ad match quality to $\bar{q}$ (i.e., whether the user is in the *Replace* arm), and $(\hat{\gamma}_i^a, \hat{\gamma}_i^q)$ are the CATEs estimated in the first step.

Third, for consumer surplus and platform demand, we allow willingness to accept to depend on aggregate participation to capture network effects. We use data from the Instagram deactivation experiment of Bursztyn et al. (2025) and regress the inverse-hyperbolic sine of willingness to accept on a measure of platform size, getting a coefficient of 1.777. To convert WTA to WTP, we multiply by 0.05, following the median gap reported by Sunstein (2020). Platform demand in each counterfactual regime is determined as a fixed point: individual participation decisions depend on counterfactual consumer surplus, while aggregate participation feeds back into consumer surplus. Armed with the equilibrium participation rate, we compute the equilibrium counterfactual consumer surplus, engagement, and profits under each monetization regime.

Finally, we aggregate outcomes across individuals to obtain counterfactual profits, time spent on the platform, consumer surplus, and platform demand. We repeat this procedure across $B = 100$ block bootstrap (user-level) resamples to get confidence intervals for the decomposition in equation (1).

5.2 Results

Figure 12 reports the mean effect of switching from the status-quo business model (personalized ad loads with microtargeting) to a subscription-based model. Estimates are averaged across $B = 100$ bootstrap replications, with percentile 95% confidence intervals shown for the total (paywall) effect. The figure decomposes this total effect into the four components described above: eliminating microtargeting, eliminating ad price differentiation, eliminating

ad-load discrimination, and replacing ads with an optimal subscription.

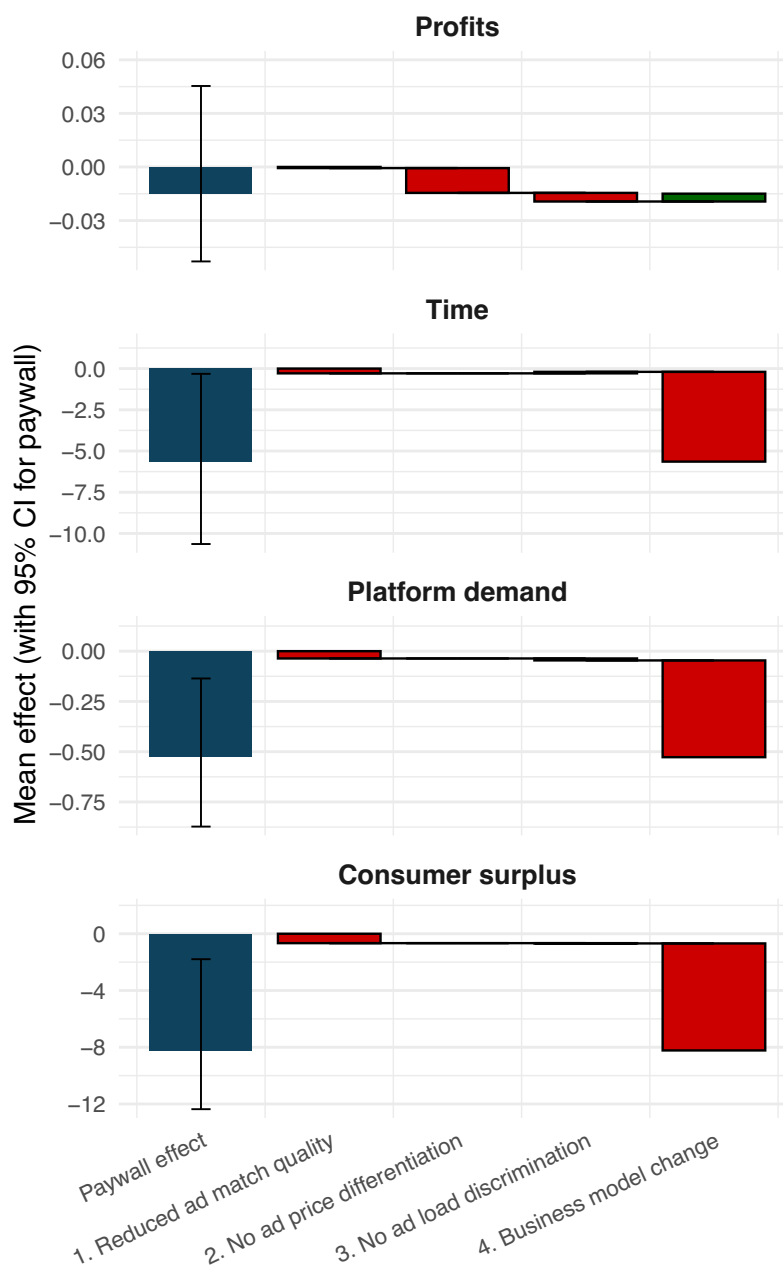


FIGURE 12: COUNTERFACTUAL MONETIZATION DECOMPOSITION

Note: The blue bars show the average treatment effect of switching from the status-quo business model (microtargeted ads with personalized ad loads and prices) to a subscription-based model, averaged across 100 bootstrap replications. Bootstrapped 95% confidence intervals are reported. The figure also decomposes this total effect into four components: (i) the change induced by eliminating microtargeting while holding each user's ad load fixed; (ii) the change from eliminating ad-price differentiation by setting all ad prices to their untargeted level; (iii) the change from eliminating ad-load discrimination by moving all users to a common ad load; and (iv) the change from replacing ads with an optimally chosen subscription fee. Positive components are shown in green, and negative components in red.

The ad-based business model performs at least as well as the subscription-based alternative across all outcomes we consider. Profits under the ad-based model are slightly higher, though not statistically distinguishable from those under subscriptions, while consumer surplus is higher under advertising.

The mechanisms differ across outcomes. For profits, the main advantage of advertising comes from ad price differentiation enabled by microtargeting—the ability to charge advertisers different rates for access to different user segments. For consumer surplus, demand, and engagement, the differences are driven primarily by the business model change itself, rather than by targeting or ad-load discrimination.

We can further decompose the fourth component in the case of profits as a term related to the relative revenue per user and the relative sensitivity to monetary prices compared to advertising loads:

$$\begin{aligned}
& \underbrace{\pi(\bar{p}, a = 0, \bar{q}, s^*) - \pi(\bar{p}, \bar{a}, \bar{q}, s = 0)}_{\text{4. Business model change}} = \underbrace{[\bar{p} \bar{a} t(\bar{a}, \bar{q}) - s^*]}_{\text{4a. Relative revenue per user}} x(\bar{a}, \bar{q}, s = 0) \\
& + s^* \underbrace{\left\{ \underbrace{[x(0, \bar{q}, s = 0) - x(0, \bar{q}, s^*)]}_{\text{4b. Sensitivity to monetary prices}} - \underbrace{[x(0, \bar{q}, s = 0) - x(\bar{a}, \bar{q}, s = 0)]}_{\text{4c. Sensitivity to ad loads}} \right\}}_{\text{Relative price sensitivity}}.
\end{aligned}$$

Appendix Figure A12 reveals that zooming in on this fourth component shows that a key driver of the relative profitability of advertising is that users are substantially more sensitive to monetary prices than to ad loads. While subscriptions generate higher revenue per participating user, this advantage is largely offset by lower participation and engagement, an effect that is amplified by network effects in platform demand.

Lastly, Appendix Figure A13 reports the results from an additional counterfactual exercise in which we consider a hybrid “opt-in” business model. Under this regime, users can choose between the status quo advertising-based version of the platform and an ad-free subscription. This option has recently been implemented by Meta in Europe, following regulatory changes that require platforms to offer users a choice between personalized advertising and a paid, ad-free alternative. The figure shows that introducing an opt-in subscription has negligible effects on profits, engagement, platform demand, and consumer surplus relative to the status quo. This result follows directly from our earlier decomposition: because users are substantially more sensitive to monetary prices than to ad loads, only a small fraction

of users (approximately 17% in our simulations) choose to subscribe to the ad-free version. As a consequence, the opt-in model leaves aggregate outcomes largely unchanged relative to a purely advertising-based business model.

Overall, these counterfactuals suggest that advertising is a relatively efficient monetization strategy on social media. It generates comparable or higher profits while delivering higher consumer surplus because users appear much more willing to pay with attention than with money. This helps rationalize the prevalence of ad-based business models on social media platforms.

6 Conclusion

As new privacy regulations are rolled out, social media will have to re-evaluate and potentially adjust their business models. In this paper, we evaluate the ad-funded business model of social media platforms under alternative data availability. Using a custom browser extension, we show that ad loads serve as prices on social media. Less personalized ads make the effective ad load “price” higher for consumers: removing targeting reduces engagement, yet does not affect platform valuation, consistent with the privacy paradox. Interestingly, platforms systematically personalize ad loads across users, with up to two-thirds of the daily variation in ad loads coming from across-user differences. This suggests that tech regulations targeting the ability of platforms to personalize ads will have large distributional effects on consumers. Counterfactual simulations suggest that an ad-based business model gives comparable short-term profits but higher engagement and consumer surplus than a subscription-based one, which helps rationalize the prevalence of ads on social media. The key mechanism is that users are much less sensitive to ad loads than to monetary prices, making advertising a relatively efficient revenue source.

References

- Acemoglu, Daron, Daniel Huttenlocher, Asuman Ozdaglar, and James Siderius (2024). *Online Business Models, Digital Ads, and User Welfare*. Tech. rep. National Bureau of Economic Research.
- Acquisti, Alessandro and Jens Grossklags (2005). “Privacy and rationality in individual decision making”. In: *IEEE security & privacy* 3.1, pp. 26–33.
- Alcobendas, Miguel, Shunto Kobayashi, Ke Shi, and Matthew Shum (2023). “The impact of privacy protection on online advertising markets”. In: *Available at SSRN 3782889*.
- Allcott, Hunt, Luca Braghieri, Sarah Eichmeyer, and Matthew Gentzkow (2020). “The welfare effects of social media”. In: *American economic review* 110.3, pp. 629–676.
- Ambrus, Attila, Emilio Calvano, and Markus Reisinger (2016). “Either or both competition: A “two-sided” theory of advertising with overlapping viewerships”. In: *American Economic Journal: Microeconomics* 8.3, pp. 189–222.
- Anderson, Simon P and Stephen Coate (2005). “Market provision of broadcasting: A welfare analysis”. In: *The Review of Economic Studies* 72.4, pp. 947–972.
- Anderson, Simon P and André De Palma (2012). “Competition for attention in the information (overload) age”. In: *The RAND Journal of Economics* 43.1, pp. 1–25.
- Anderson, Simon P. and Martin Peitz (May 2023). “Ad Clutter, Time Use, and Media Diversity”. In: *American Economic Journal: Microeconomics* 15.2, pp. 227–70. DOI: [10.1257/mic.20210139](https://doi.org/10.1257/mic.20210139). URL: <https://www.aeaweb.org/articles?id=10.1257/mic.20210139>.
- Argentesi, Elena, Paolo Buccirossi, Emilio Calvano, Tomaso Duso, Alessia Marrazzo, and Salvatore Nava (2021). “Merger policy in digital markets: An ex post assessment”. In: *Journal of Competition Law & Economics* 17.1, pp. 95–140.
- Aridor, Guy (2025). “Measuring substitution patterns in the attention economy: An experimental approach”. In: *The RAND Journal of Economics* 56.3, pp. 302–324.
- Aridor, Guy, Yeon-Koo Che, and Tobias Salz (2023). “The Effect of Privacy Regulation on the Data Industry: Empirical Evidence from GDPR”. In: *RAND Journal of Economics* 54.4, pp. 695–730. DOI: [10.1111/1756-2171.12455](https://doi.org/10.1111/1756-2171.12455). URL: <https://doi.org/10.1111/1756-2171.12455>.
- Aridor, Guy, Yeon-Koo Che, Brett Hollenbeck, Maximilian Kaiser, and Daniel McCarthy (2025a). “Evaluating the impact of privacy regulation on e-commerce firms: Evidence from apple’s app tracking transparency”. In: *Management Science*.
- Aridor, Guy, Rafael Jiménez-Durán, Ro’ee Levy, and Lena Song (2025b). “Experiments on social media”. In: *Handbook of Experimental Methods in the Social Sciences*. Edward Elgar Publishing.
- Aridor, Guy, Rafael Jiménez-Durán, Ro’ee Levy, and Lena Song (2024). “The economics of social media”. In: *Journal of Economic Literature* 62.4, pp. 1422–1474.
- Athey, Susan, Emilio Calvano, and Joshua S Gans (2018). “The impact of consumer multi-homing on advertising markets and media competition”. In: *Management science* 64.4, pp. 1574–1590.

- Athey, Susan, Julie Tibshirani, Stefan Wager, et al. (2019). “Generalized random forests”. In: *Annals of Statistics* 47.2, pp. 1148–1178.
- Beknazar-Yuzbashev, George, Rafael Jiménez-Durán, Jesse McCrosky, and Mateusz Stalinski (2025). “Toxic content and user engagement on social media: Evidence from a field experiment”. In.
- Beknazar-Yuzbashev, George, Rafael Jiménez-Durán, and Mateusz Stalinski (2024). “A model of harmful yet engaging content on social media”. In: *AEA Papers and Proceedings*. Vol. 114. American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203, pp. 678–683.
- Brynjolfsson, Erik, Avinash Collis, Daniel Deisenroth, Haritz Garro, Daley Kutzman, Asad Liaqat, and Nils Wernerfelt (Dec. 2025). “The Consumer Welfare Effects of Online Ads: Evidence from a Nine-Year Experiment”. In: *American Economic Review: Insights* 7.4, pp. 447–62. DOI: [10.1257/aeri.20240452](https://doi.org/10.1257/aeri.20240452). URL: <https://www.aeaweb.org/articles?id=10.1257/aeri.20240452>.
- Bursztyń, Leonardo, Benjamin Handel, Rafael Jiménez-Durán, and Christopher Roth (Dec. 2025). “When Product Markets Become Collective Traps: The Case of Social Media”. In: *American Economic Review* 115.12, pp. 4105–36. DOI: [10.1257/aer.20231468](https://doi.org/10.1257/aer.20231468). URL: <https://www.aeaweb.org/articles?id=10.1257/aer.20231468>.
- Calvano, Emilio and Michele Polo (2020). “Strategic differentiation by business models: Free-to-air and pay-TV”. In: *The Economic Journal* 130.625, pp. 50–64.
- (2021). “Market power, competition and innovation in digital markets: A survey”. In: *Information Economics and Policy* 54, p. 100853.
- Cameron, A Colin and Douglas L Miller (2015). “A practitioner’s guide to cluster-robust inference”. In: *Journal of human resources* 50.2, pp. 317–372.
- Candogan, Ozan, Kostas Bimpikis, and Asuman Ozdaglar (2012). “Optimal pricing in networks with externalities”. In: *Operations Research* 60.4, pp. 883–905.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer (2019). “The effect of minimum wages on low-wage jobs”. In: *The Quarterly Journal of Economics* 134.3, pp. 1405–1454.
- Center, Stigler (2019). *Stigler Committee on Digital Platforms*. Stigler Center.
- Driscoll, John C and Aart C Kraay (1998). “Consistent covariance matrix estimation with spatially dependent panel data”. In: *Review of economics and statistics* 80.4, pp. 549–560.
- Dubé, Jean-Pierre, John G Lynch, Dirk Bergemann, Mert Demirer, Avi Goldfarb, Garrett Johnson, Anja Lambrecht, Tesary Lin, Anna Tuchman, and Catherine E Tucker (2025). “The intended and unintended consequences of privacy regulation for consumer marketing”. In.
- European Commission (2025). *Commission finds Apple and Meta in breach of the Digital Markets Act*. https://ec.europa.eu/commission/presscorner/detail/en/ip_25_1085. Accessed: 2026-01-14.

- Farronato, Chiara, Andrey Fradkin, and Chris Karr (2024). *Webmunk: A new tool for studying online behavior and digital platforms*. Tech. rep. National Bureau of Economic Research.
- Gentzkow, Matthew, Jesse M Shapiro, Frank Yang, and Ali Yurukoglu (2024). “Pricing power in advertising markets: Theory and evidence”. In: *American Economic Review* 114.2, pp. 500–533.
- Goldfarb, Avi and Verina F Que (2023). “The economics of digital privacy”. In: *Annual Review of Economics* 15.1, pp. 267–286.
- Goldfarb, Avi and Catherine Tucker (2012). “Shifts in privacy concerns”. In: *American Economic Review* 102.3, pp. 349–353.
- Goldfarb, Avi and Catherine E Tucker (2011). “Privacy regulation and online advertising”. In: *Management science* 57.1, pp. 57–71.
- Goli, Ali, Jason Huang, David Reiley, and Nickolai M Riabov (2025a). “Measuring consumer sensitivity to audio advertising: a long-run field experiment on Pandora internet radio”. In: *Quantitative Marketing and Economics*, pp. 1–31.
- Goli, Ali, David H Reiley, and Hongkai Zhang (2025b). “Personalizing ad load to optimize subscription and ad revenues: Product strategies constructed from experiments on pandora”. In: *Marketing Science* 44.2, pp. 327–352.
- Goodman, Joseph, Lancelot Henry de Frahan, Justin E. Holz, John A. List, Niall MacMenamin, Evan C. McKay, Magne Mogstad, Sally Sadoff, and Hal Sider (Jan. 2026). *Consumer Demand and Market Competition with Time-Intensive Goods*. Working Paper 34743. National Bureau of Economic Research. DOI: [10.3386/w34743](https://doi.org/10.3386/w34743). URL: <https://www.nber.org/papers/w34743>.
- Hoechle, Daniel (2007). “Robust standard errors for panel regressions with cross-sectional dependence”. In: *The stata journal* 7.3, pp. 281–312.
- Johnson, Garrett A (2024). “4. Economic Research on Privacy Regulation: Lessons from the GDPR and Beyond”. In: *The Economics of Privacy*. University of Chicago Press, pp. 97–126.
- Katz, Justin and Hunt Allcott (2025). *Digital Media Mergers: Theory and Application to Facebook-Instagram*. Tech. rep. Working paper.
- Kling, Jeffrey R., Jeffrey B. Liebman, and Lawrence F. Katz (2007). “Experimental Analysis of Neighborhood Effects”. In: *Econometrica* 75.1, pp. 83–119.
- Lin, Tesary (2022). “Valuing intrinsic and instrumental preferences for privacy”. In: *Marketing Science* 41.4, pp. 663–681.
- Liu, Yi, Pinar Yildirim, and Z John Zhang (2022). “Implications of revenue models and technology for content moderation strategies”. In: *Marketing Science* 41.4, pp. 831–847.
- Moshary, Sarah (2021). “Sponsored search in equilibrium: evidence from two experiments”. In: *Available at SSRN 3903602*.
- Pigou, Arthur (1920). *The economics of welfare*. Routledge.
- Sunstein, Cass R (2020). “Valuing facebook”. In: *Behavioural Public Policy* 4.3, pp. 370–381.

- Wernerfelt, Nils, Anna Tuchman, Bradley T Shapiro, and Robert Moakler (2025). “Estimating the value of offsite tracking data to advertisers: Evidence from meta”. In: *Marketing Science* 44.2, pp. 268–286.
- Wilbur, Kenneth C (2008). “A two-sided, empirical model of television advertising and viewing markets”. In: *Marketing science* 27.3, pp. 356–378.

A Figures

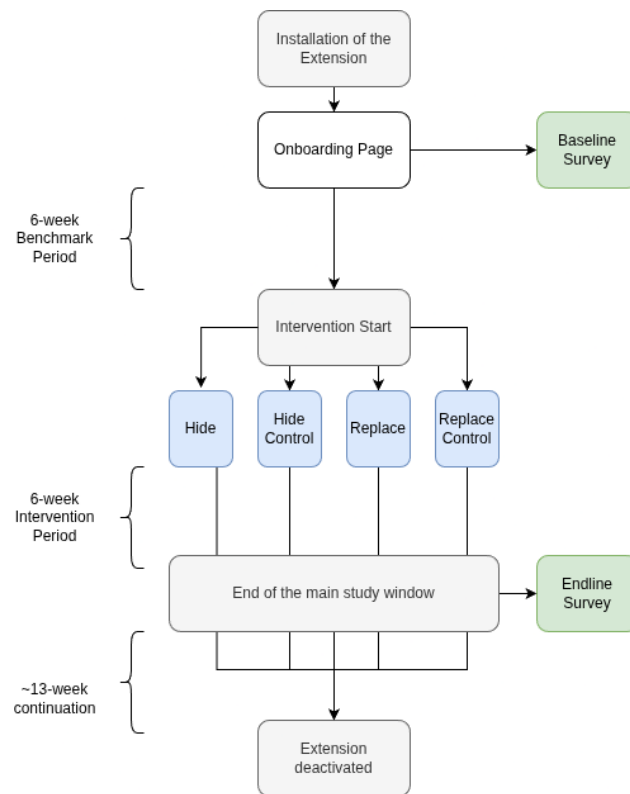
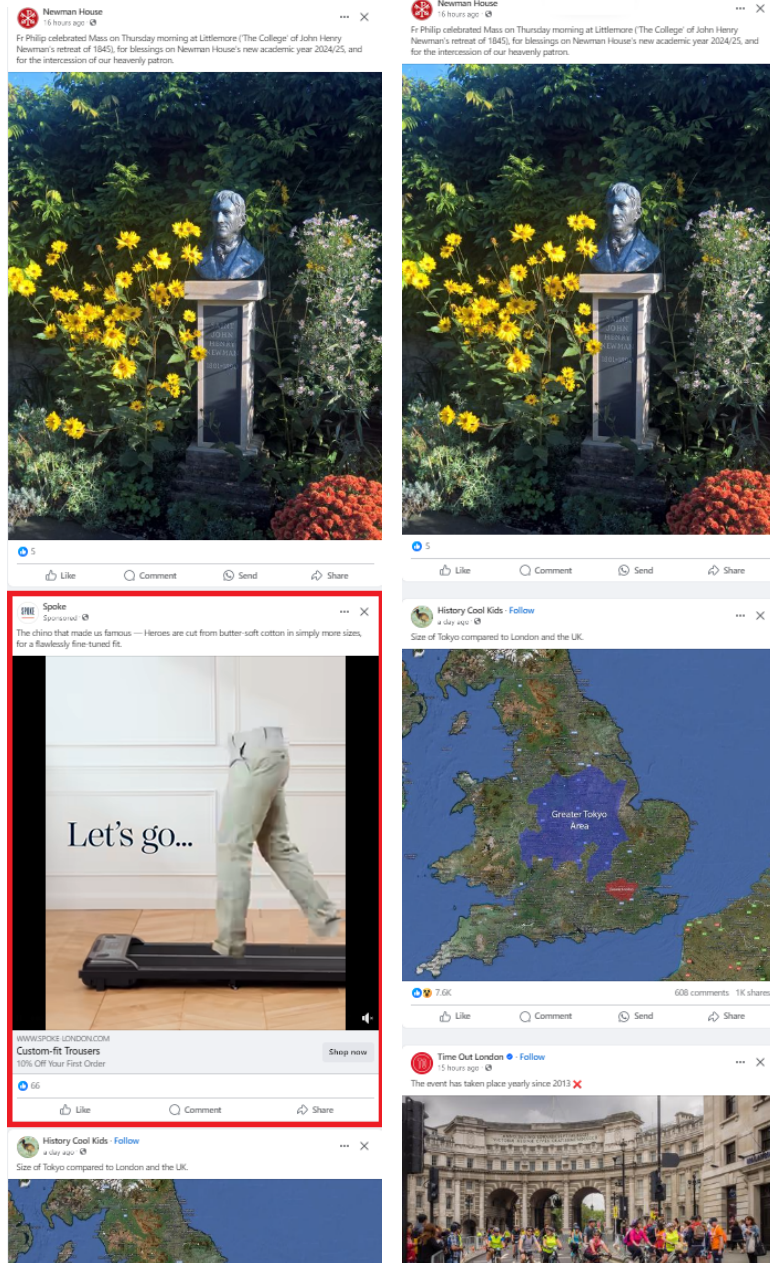


FIGURE A1: STUDY FLOW



(a) Original Feed

(b) Moderated Feed

FIGURE A2: HIDING INTERVENTION

Note: The **left panel** shows an unmoderated feed with a non-ad post, followed by an ad from *Spoke*, a company offering custom-fit garments, and another non-ad post. The ad is framed in red for demonstration purposes. In the **right panel**, we display the moderated feed after the ad-hiding action has been applied. The *Spoke* ad is removed, and the second non-ad post is pulled upward, along with the content below it. As a result, a third non-ad post, from *Time Out London*, becomes visible.



(a) Replacement Ad



(b) Possible Interactions

FIGURE A3: REPLACEMENT INTERVENTION

Note: The **left panel** of the figure shows an example of a replacement ad used in the replacement intervention. The **right panel** illustrates potential interactions with the replacement ad. Users can like the ad or react to it using any available emoji, and their reaction will be displayed naturally, as shown in the figure. If someone attempts to comment on or share the ad, a small pop-up appears, informing them that commenting and sharing are disabled (as depicted in the figure).

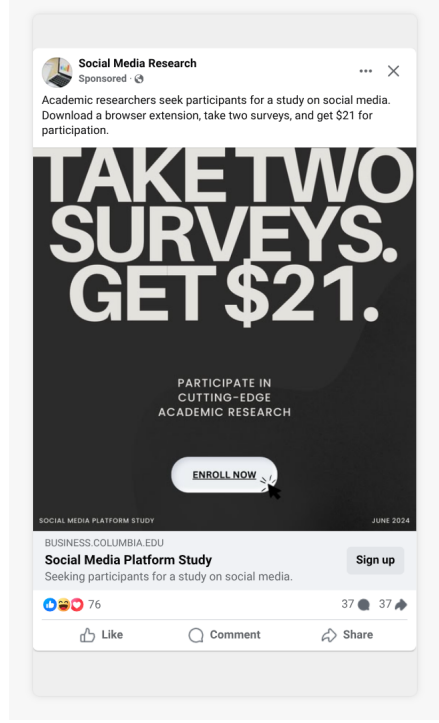


FIGURE A4: EXAMPLE OF A RECRUITMENT AD

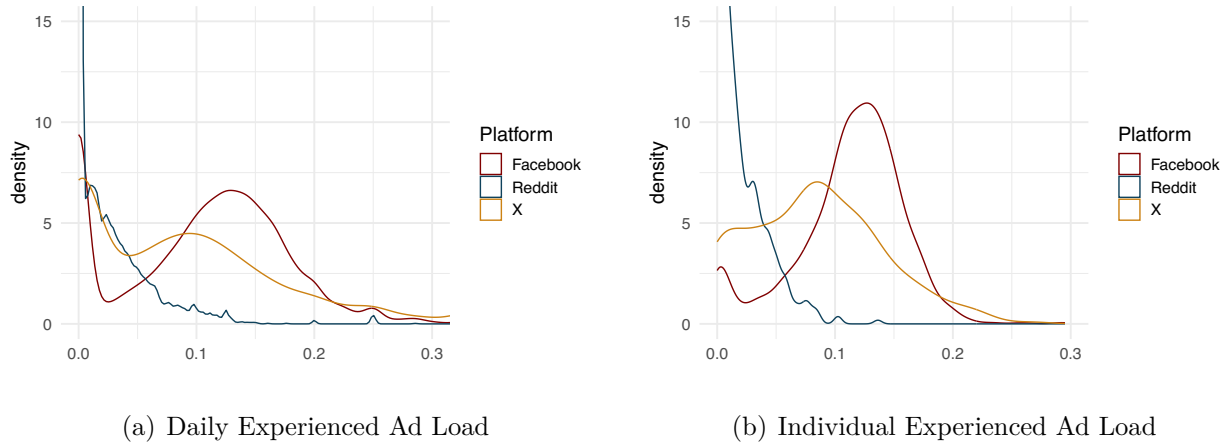


FIGURE A5: DISTRIBUTION OF AD LOAD BY PLATFORM

Note: **Panel A** presents the distributions of the experienced advertising load per person-day on Facebook, Reddit, and X, based on the baseline period data. **Panel B** presents the equivalent distributions per person. The ad load is defined as the number of ads as a share of total posts and comments shown to the user on each platform during periods outside of any treatment. The x-axis for Panel A is truncated 0.3 and the y-axis is truncated at 30. Panel B relies on the subsample of users who viewed at least 50 posts and comments throughout the relevant period.

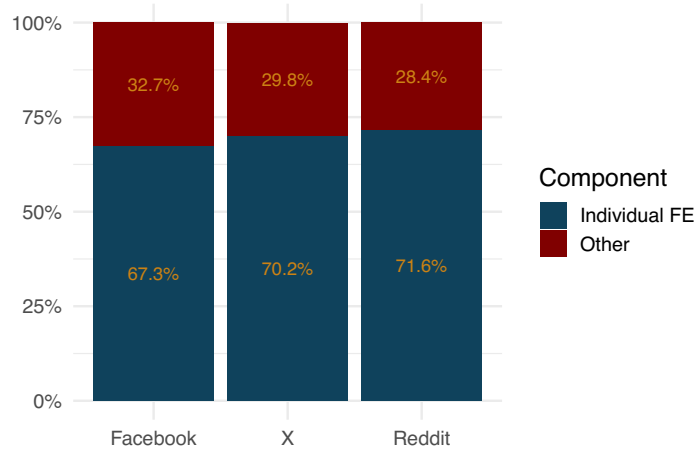


FIGURE A6: AD LOADS: VARIANCE DECOMPOSITION

Note: The figure reports the results of a variance decomposition exercise, showing the share of variance in feed ad load attributable to individual fixed effects for each social media platform (Facebook, Reddit, and X). Estimates are obtained from a fixed effects regression with individual fixed effects. The figure is based on untreated observations with at least 50 feed posts and comments observed for each day.

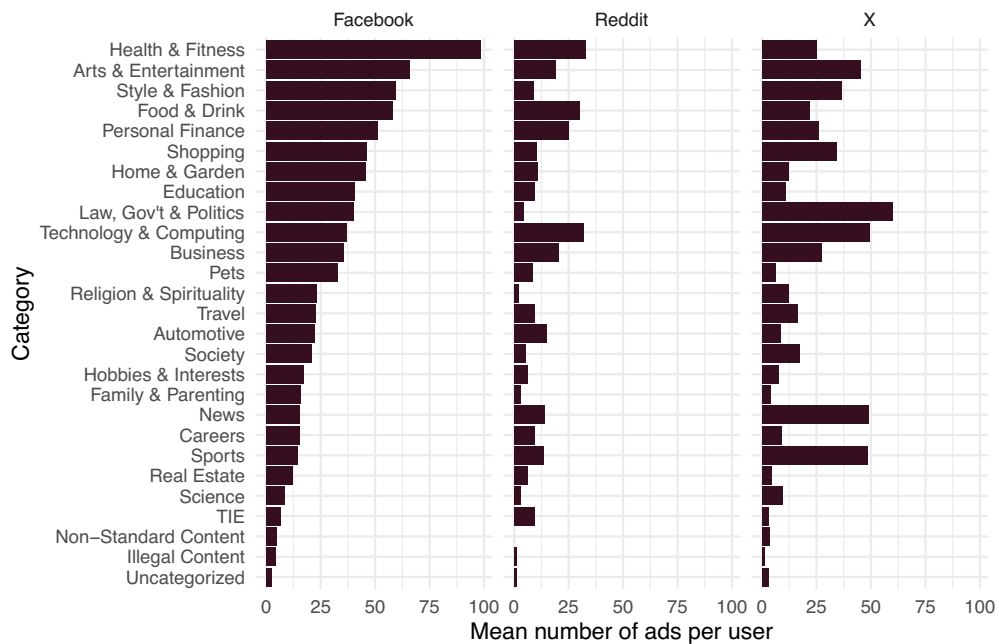


FIGURE A7: AD CATEGORIES BY PLATFORM

Note: The figure reports the results the mean number of ads by topic category per user, separately for Facebook, Reddit, and X. The categories are sorted by their frequency in our Facebook data. The categorization was performed using generative AI. The computations are based on subsamples of people who watched at least 50 ads on each of the relevant platforms.

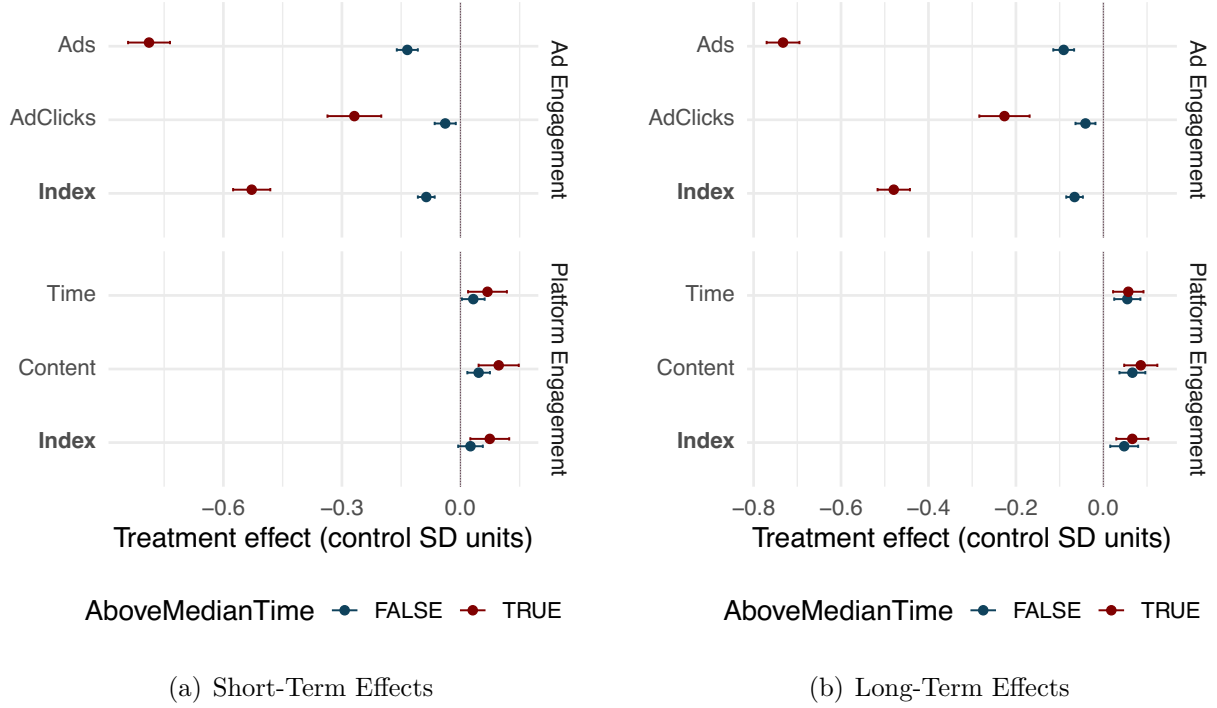


FIGURE A8: HETEROGENEITY OF ENGAGEMENT EFFECTS: HIDING (ASINH)

Note: The figure displays estimated treatment effects of the hiding intervention on Facebook relative to *Hide Control* separately for users whose baseline active time on Facebook was above and below the median. The estimation is based on Equation 4, with $\beta + \phi$ representing the above-median coefficient and β representing the below-median coefficient. The regression relies on the main experimental sample. The dependent variables are: (i) number of ad impressions, (ii) number of ad clicks, (iii) an ad engagement index based on (i) and (ii), (iv) active time spent on the platform, (v) number of posts and comments shown to the user, and (vi) a platform engagement index based on (iv) and (v). **Panel A** shows results based on the six-week intervention period, whereas **Panel B** shows results based on a longer intervention period (134 day-periods or 19.1 weeks). All outcomes are transformed using the inverse hyperbolic sine (asinh) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. We include 95% confidence intervals based on Driscoll-Kraay standard errors. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day.

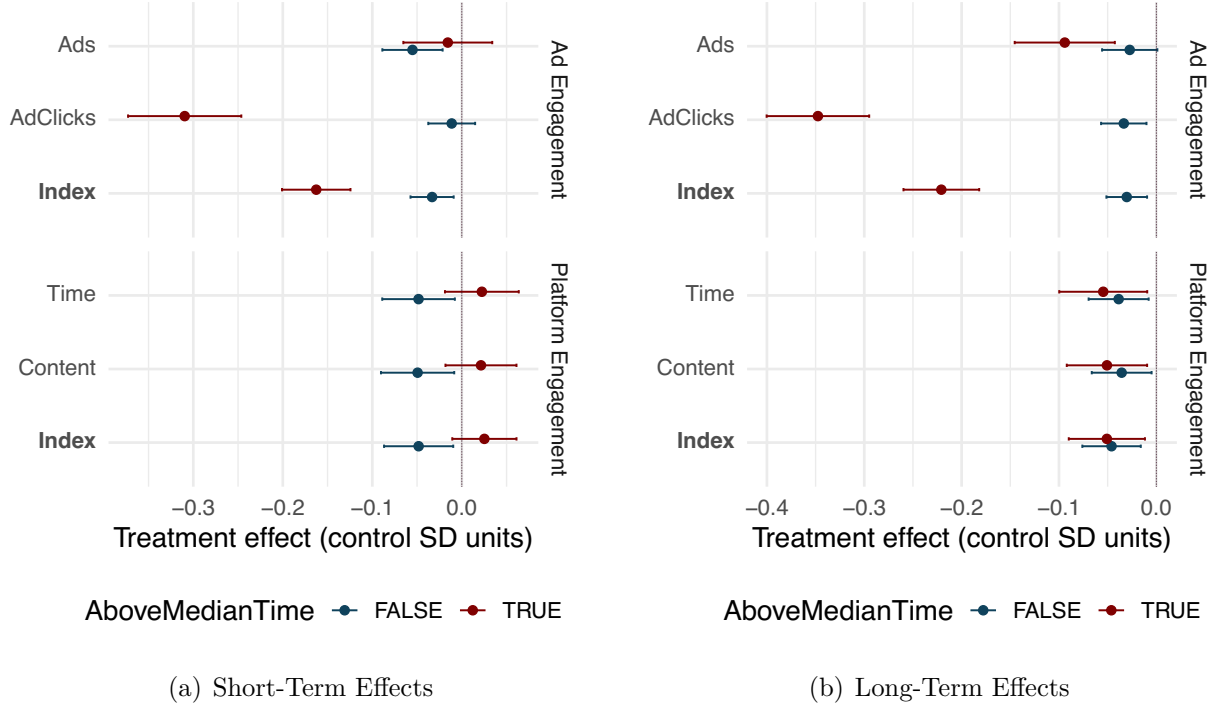


FIGURE A9: HETEROGENEITY OF ENGAGEMENT EFFECTS: REPLACEMENT (ASINH)

Note: The figure displays estimated treatment effects of the replacement intervention on Facebook relative to *Replace Control* separately for users whose baseline active time on Facebook was above and below the median. The estimation is based on Equation 4, with $\beta + \phi$ representing the above-median coefficient and β representing the below-median coefficient. The regression relies on the main experimental sample. The dependent variables are: (i) number of ad impressions, (ii) number of ad clicks, (iii) an ad engagement index based on (i) and (ii), (iv) active time spent on the platform, (v) number of posts and comments shown to the user, and (vi) a platform engagement index based on (iv) and (v). **Panel A** shows results based on the six-week intervention period, whereas **Panel B** shows results based on a longer intervention period (134 day-periods or 19.1 weeks). All outcomes are transformed using the inverse hyperbolic sine (*asinh*) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. We include 95% confidence intervals based on Driscoll-Kraay standard errors. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day.

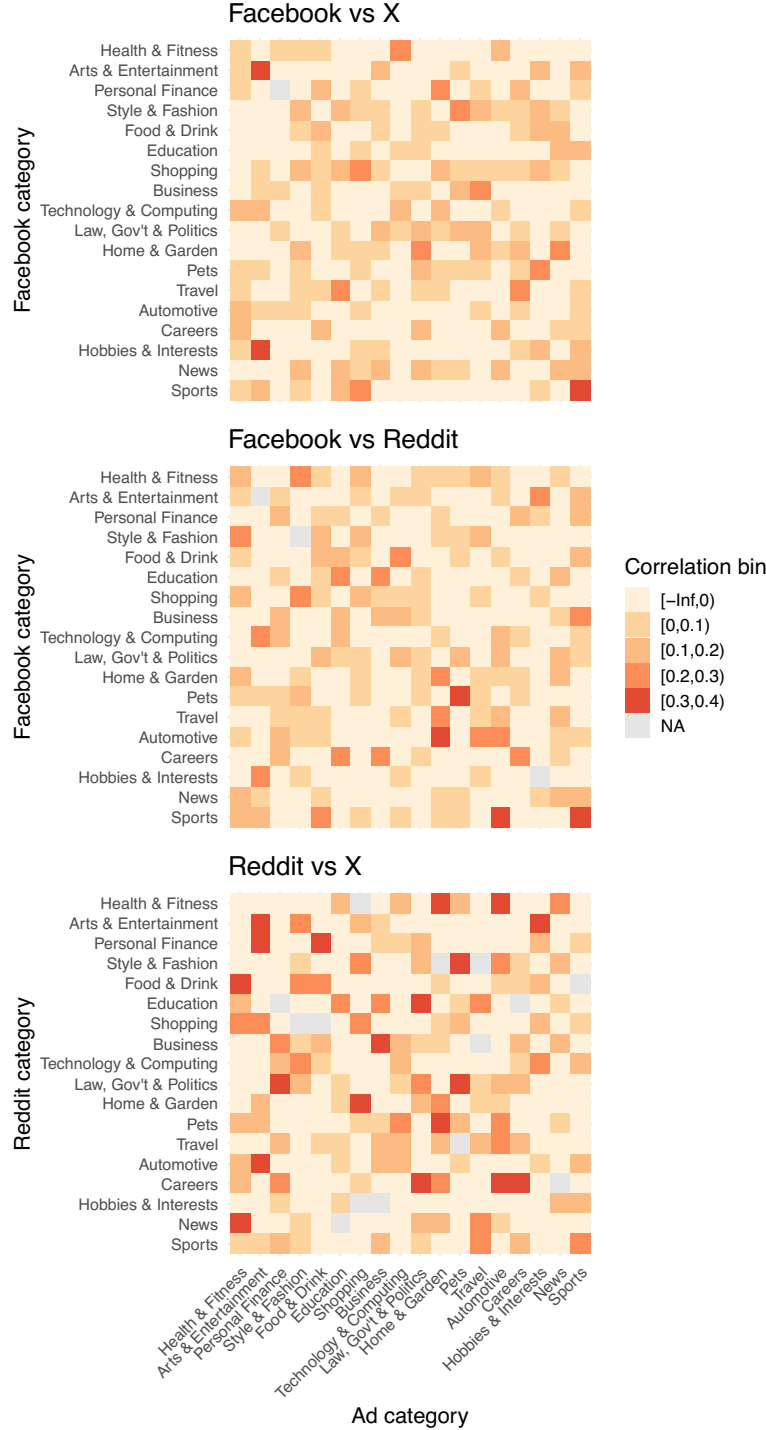


FIGURE A10: CORRELATIONS IN AD CATEGORIES BETWEEN PLATFORMS

Note: This figure presents heatmaps of correlations in topic-based ad targeting across platforms. Ads are first classified into topic categories from the IAB Content Taxonomy (see Section 4.1.3). For each user and platform, we compute the share of ads shown in each topic category and divide by the platform-level category share to obtain relative topic shares. Each cell reports the correlation between relative topic shares for a given pair of categories across two platforms. Lighter colors indicate lower correlations and darker colors indicate higher correlations. The computations are based on subsamples of people who watched at least 50 ads on each of the relevant platforms, and we limit the comparison to ad categories that appear at least 300 times on each of the platforms.

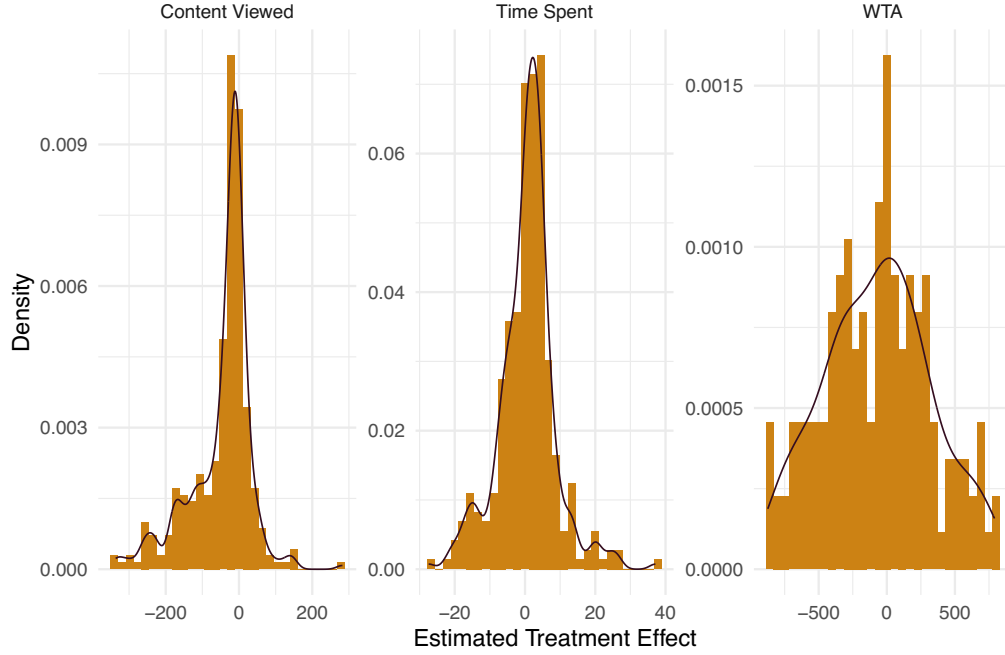


FIGURE A11: PREDICTED CATEs UNDER AD HIDING

Note: This figure shows the distribution of predicted conditional average treatment effects (CATEs) under an ad-hiding counterfactual for content viewed, time spent on the platform, and willingness to accept (WTA) to deactivate the platform. CATEs are estimated using randomized variation from the experiment. The causal effect of ad load is estimated using an instrumental forest, where changes in ad load are instrumented by assignment to the Hide versus Hide Control groups; the causal effect of ad targeting is estimated using a causal forest based on assignment to the Replace versus Replace Control groups. The feature set includes baseline demographics, baseline platform usage and ad exposure, and indicators for browser extension errors. CATEs are estimated in-sample for the relevant comparison groups and predicted out-of-sample for other treatment groups, pooling predictions across bootstrap resamples.

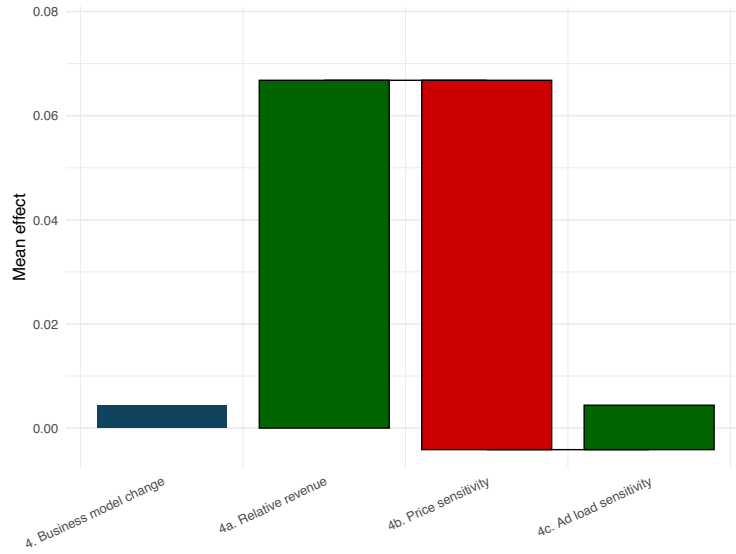


FIGURE A12: DECOMPOSITION OF THE BUSINESS MODEL CHANGE CHANNEL

Note: This figure decomposes the business-model change (blue) into relative revenue, price sensitivity, and ad-load sensitivity components. Mean effects across 100 bootstrap replications are shown. Positive components are shown in green, and negative components in red.

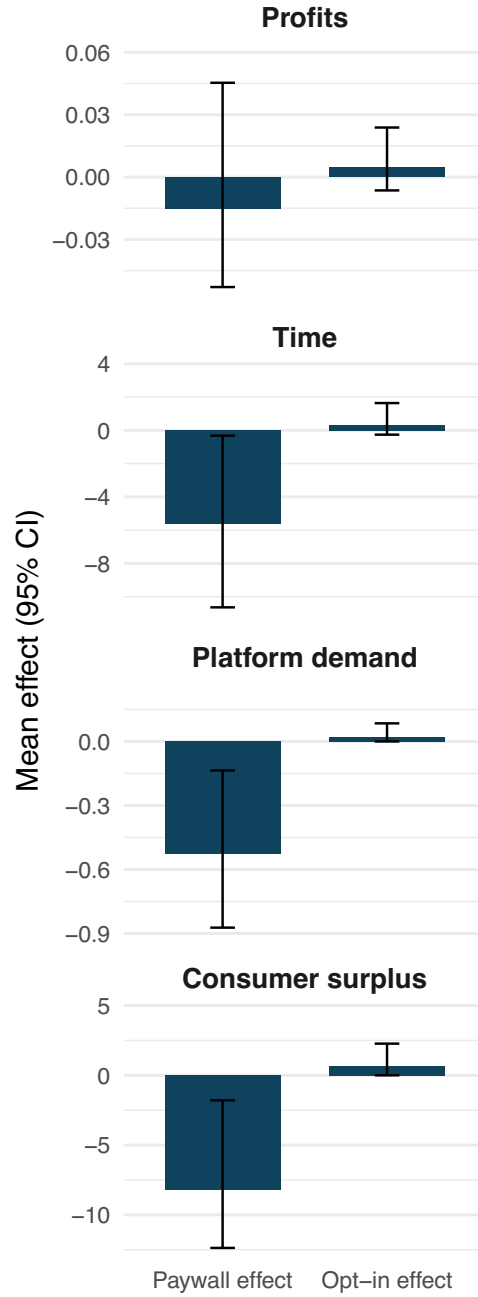


FIGURE A13: COMPARISON OF PAYWALL AND OPT-IN EFFECTS

Note: This figure compares the status-quo advertising-based business model to 1) a subscription-based model and 2) a hybrid “opt-in” model in which users may choose between the ad-supported platform and an ad-free subscription. Bars report mean effects across 100 bootstrap replications, with 95% confidence intervals.

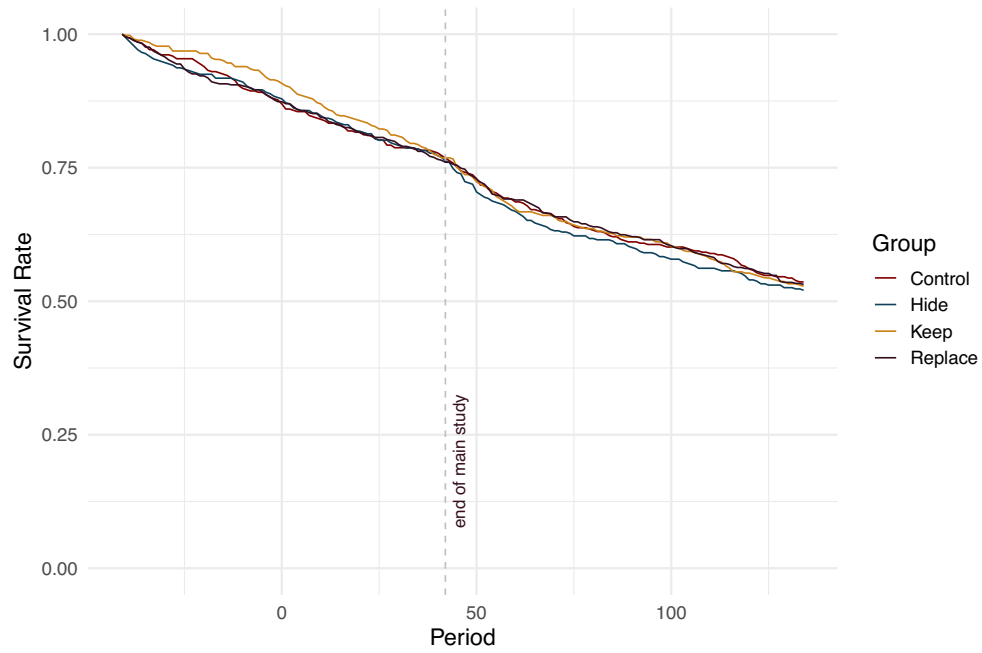


FIGURE A14: SURVIVAL RATE IN THE STUDY

Note: The figure presents the share of users enrolled in the study who survived until a particular study period, measured as the number of days relative to the intervention start date (which starts in period 1). We say that a particular user survived until period t if they were active on their browser on the day corresponding to period t or later. The survival shares are split by treatment conditions.

B Tables

TABLE B1: BALANCE TABLE: HIDE VS. HIDE CONTROL

	Control (N=414)		Treatment (N=413)		Diff. in Means	p
	Mean	Std. Dev.	Mean	Std. Dev.		
Age	43.9	13.0	44.6	13.4	0.7	0.446
Male	0.4	0.5	0.3	0.5	0.0	0.829
White	0.7	0.5	0.7	0.5	0.0	0.913
Bachelors	0.4	0.5	0.3	0.5	0.0	0.261
Income 50k+	0.4	0.5	0.4	0.5	0.0	0.399
Democrat	0.6	0.5	0.6	0.5	0.0	0.926
Facebook Desktop Usage	52.0	23.2	49.4	23.1	-2.6	0.121
Baseline Time on FB	14.3	22.5	15.8	34.4	1.4	0.474
Baseline Content on FB	73.4	132.7	80.9	185.0	7.4	0.507
Baseline Ads on FB	8.6	17.2	8.1	16.6	-0.4	0.715
Baseline Ad Clicks on FB	0.2	0.4	0.1	0.3	0.0	0.191

Note: This table compares user characteristics between the Hide treatment group and the Hide Control group in the main experimental sample. It reports group means, standard deviations, differences in means, and p-values from two-sided tests of differences in means. The variables are reported in the following order: (1) age (in years), (2) an indicator for male, (3) an indicator for identifying as white/Caucasian, (4) an indicator for having at least a bachelor's degree, (5) an indicator for having household income above \$50,000, (6) an indicator for being a Democrat, (7) desktop share of Facebook usage (from the baseline survey), (8) number of Facebook friends (from the baseline survey), (9) baseline active time on Facebook (minutes per day), (10) baseline number of posts and comments consumed per day on Facebook, (11) baseline number of ad impressions per day on Facebook, and (12) baseline number of ad clicks per day on Facebook. The p-values are based on regressions estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day.

TABLE B2: BALANCE TABLE: REPLACE VS. REPLACE CONTROL

	Control (N=445)		Treatment (N=538)		Diff. in Means	p
	Mean	Std. Dev.	Mean	Std. Dev.		
Age	44.0	12.8	45.5	13.2	1.5	0.073
Male	0.3	0.5	0.3	0.5	0.0	0.374
White	0.7	0.4	0.8	0.4	0.0	0.756
Bachelors	0.4	0.5	0.4	0.5	0.0	0.414
Income 50k+	0.4	0.5	0.4	0.5	0.0	0.560
Democrat	0.6	0.5	0.6	0.5	0.0	0.428
Facebook Desktop Usage	50.8	22.7	51.3	24.1	0.5	0.764
Baseline Time on FB	15.7	27.3	15.6	29.6	-0.1	0.961
Baseline Content on FB	83.3	180.9	82.3	165.1	-1.0	0.927
Baseline Ads on FB	9.8	24.1	8.9	19.3	-0.9	0.524
Baseline Ad Clicks on FB	0.1	0.5	0.2	1.0	0.1	0.253

Note: This table compares user characteristics between the Hide treatment group and the Hide Control group in the main experimental sample. It reports group means, standard deviations, differences in means, and p-values from two-sided tests of differences in means. The variables are reported in the following order: (1) age (in years), (2) an indicator for male, (3) an indicator for identifying as white/Caucasian, (4) an indicator for having at least a bachelor's degree, (5) an indicator for having household income above \$50,000, (6) an indicator for being a Democrat, (7) desktop share of Facebook usage (from the baseline survey), (8) number of Facebook friends (from the baseline survey), (9) baseline active time on Facebook (minutes per day), (10) baseline number of posts and comments consumed per day on Facebook, (11) baseline number of ad impressions per day on Facebook, and (12) baseline number of ad clicks per day on Facebook. The p-values are based on regressions estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day.

TABLE B3: SHORT-TERM EFFECTS OF INTERVENTIONS ON ADVERTISING LOADS

	User-Side Outcomes		Platform-Side Outcomes	
	Experienced Ad Load	Experienced Advertiser Distance	Supplied Ad Load	Supplied Advertiser Distance
<i>Panel A: Hide vs Pure Control</i>				
Treated	-0.113*** (0.003)	-0.070*** (0.011)	0.003 (0.002)	-0.016* (0.008)
Mean	0.137	0.997	0.137	0.997
SD	0.113	0.233	0.113	0.233
N	27334	16736	27335	18999
<i>Panel B: Replace vs Replace Control</i>				
Treated	0.000 (0.004)	0.109*** (0.007)	-0.000 (0.004)	0.015** (0.006)
Mean	0.137	0.999	0.137	0.999
SD	0.124	0.233	0.124	0.233
N	32344	21273	32344	21257

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). The dependent variables are: (i) experienced ad load, advertisement as a share of posts and comments viewed by the user, (ii) experienced advertising distance, (iii) supplied ad load, advertisement as a share of posts and comments that a user was targeted with, before any filtering was applied, and (iv) supplied advertiser distance, based on ads that the user was targeted with, before any filtering was applied. Ad distance is defined as the Euclidean distance between a user's characteristics and the average demographic profile targeted by that advertiser, based on baseline data. The unit of observation is the individual-day, where day is measured relative to the intervention date. Driscoll-Kraay standard errors are parenthesized. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B4: LONG-TERM EFFECTS OF INTERVENTIONS ON ADVERTISING LOADS

	User-Side Outcomes		Platform-Side Outcomes	
	Experienced Ad Load	Experienced Advertiser Distance	Supplied Ad Load	Supplied Advertiser Distance
<i>Panel A: Hide vs Pure Control</i>				
Treated	-0.117*** (0.003)	-0.065*** (0.009)	0.002 (0.002)	-0.021*** (0.008)
Mean	0.142	1.00	0.142	1.00
SD	0.12	0.237	0.12	0.237
N	45751	23724	45752	29278
<i>Panel B: Replace vs Replace Control</i>				
Treated	0.001 (0.003)	0.100*** (0.006)	0.000 (0.003)	0.012** (0.005)
Mean	0.143	1.00	0.143	1.00
SD	0.127	0.235	0.127	0.235
N	53674	32190	53674	32109

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). The dependent variables are: (i) experienced ad load, advertisement as a share of posts and comments viewed by the user, (ii) experienced advertising distance, (iii) supplied ad load, advertisement as a share of posts and comments that a user was targeted with, before any filtering was applied, and (iv) supplied advertiser distance, based on ads that the user was targeted with, before any filtering was applied. Ad distance is defined as the Euclidean distance between a user's characteristics and the average demographic profile targeted by that advertiser, based on baseline data. The unit of observation is the individual-day, where day is measured relative to the intervention date. Driscoll-Kraay standard errors are parenthesized. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B5: SHORT-TERM EFFECTS OF INTERVENTIONS ON ENGAGEMENT

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.026 (0.017)	0.646 (0.576)	5.641* (3.282)	-0.192*** (0.015)	-7.049*** (0.334)	-0.038*** (0.011)
Mean	0.00	11.26	59.93	0.00	7.44	0.066
SD	0.85	28.08	175.10	0.808	25.28	0.357
N	69468	69468	69468	69468	69468	69468
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.011 (0.011)	-0.674 (0.475)	-3.738 (2.797)	-0.150*** (0.014)	-1.586*** (0.450)	-0.093*** (0.009)
Mean	0.00	12.37	64.52	0.00	8.30	0.067
SD	0.852	31.74	204.40	0.796	32.80	0.37
N	82572	82572	82572	82572	82572	82572

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B6: LONG-TERM EFFECTS OF INTERVENTIONS ON ENGAGEMENT

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.020* (0.012)	0.572 (0.420)	6.632*** (2.476)	-0.167*** (0.011)	-6.397*** (0.326)	-0.039*** (0.008)
Mean	0.00	11.90	63.08	0.00	7.89	0.073
SD	0.85	30.16	180.03	0.813	26.50	0.422
N	116515	116515	116515	116515	116515	116515
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.032** (0.013)	-1.701*** (0.467)	-9.401*** (3.073)	-0.178*** (0.016)	-2.033*** (0.457)	-0.109*** (0.011)
Mean	0.00	13.09	69.67	0.00	8.90	0.067
SD	0.864	33.66	203.49	0.808	30.11	0.38
N	139275	139275	139275	139275	139275	139275

Note: **Panel A** presents the estimated treatment effects of the hiding intervention on Facebook, relative to the Hide Control group, based on Equation 3. **Panel B** reports the corresponding estimates for the replacement intervention, relative to the Replace Control group. All regressions are based on the extended sample, which involves 19.1 weeks of the intervention period (long-term effects) as opposed to 6 weeks specified in the pre-registration (short-term effects). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B7: SHORT-TERM ENGAGEMENT EFFECTS (ASINH)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.066*** (0.021)	0.117*** (0.038)	0.202*** (0.051)	-0.285*** (0.013)	-0.687*** (0.029)	-0.033*** (0.004)
Mean	0.00	1.20	1.63	0.00	0.904	0.05
SD	0.976	1.82	2.48	0.841	1.59	0.241
N	69468	69468	69468	69468	69468	69468
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.021* (0.013)	-0.043 (0.026)	-0.069** (0.035)	-0.087*** (0.012)	-0.048** (0.022)	-0.035*** (0.004)
Mean	0.00	1.22	1.64	0.00	0.883	0.05
SD	0.978	1.85	2.49	0.844	1.59	0.242
N	82572	82572	82572	82572	82572	82572

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. All outcomes are transformed using the inverse hyperbolic sine (asinh) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B8: LONG-TERM ENGAGEMENT EFFECTS (ASINH)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.054*** (0.017)	0.103*** (0.029)	0.175*** (0.042)	-0.260*** (0.010)	-0.640*** (0.023)	-0.032*** (0.004)
Mean	0.00	1.22	1.72	0.00	0.951	0.053
SD	0.976	1.84	2.52	0.841	1.62	0.256
N	116515	116515	116515	116515	116515	116515
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.057*** (0.012)	-0.107*** (0.023)	-0.143*** (0.029)	-0.110*** (0.013)	-0.089*** (0.023)	-0.040*** (0.004)
Mean	0.00	1.25	1.74	0.00	0.959	0.05
SD	0.978	1.87	2.53	0.839	1.64	0.243
N	139275	139275	139275	139275	139275	139275

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). All regressions are based on the extended sample, which involves 19.1 weeks of the intervention period (long-term effects) as opposed to 6 weeks specified in the pre-registration (short-term effects). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. All outcomes are transformed using the inverse hyperbolic sine (asinh) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B9: SHORT-TERM ENGAGEMENT EFFECTS (TRIMMING)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.057*** (0.018)	1.604*** (0.559)	11.155*** (3.233)	-0.277*** (0.013)	-6.190*** (0.335)	-0.055*** (0.005)
Mean	0.00	10.90	54.47	0.00	6.34	0.051
SD	0.86	27.43	151.92	0.811	18.19	0.297
N	67956	68712	68712	67956	68712	68712
<i>Panel B: Replace vs Replace Control</i>						
Treated	0.007 (0.012)	-0.275 (0.509)	-3.326 (2.466)	-0.081*** (0.017)	-1.140** (0.434)	-0.037*** (0.007)
Mean	0.00	11.46	54.47	0.00	6.62	0.064
SD	0.854	29.64	159.06	0.83	22.31	0.352
N	80892	81732	81732	80892	81732	81732

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). All regressions are based on the sample excluding the top 1 percentile of observations—the exclusion is performed for each outcome separately. The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B10: LONG-TERM ENGAGEMENT EFFECTS (TRIMMING)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.047*** (0.013)	1.631*** (0.385)	10.893*** (2.282)	-0.235*** (0.010)	-5.468*** (0.258)	-0.049*** (0.004)
Mean	0.00	11.30	56.10	0.00	6.57	0.052
SD	0.855	28.93	152.01	0.819	18.74	0.307
N	113783	115231	115111	113765	115165	115169
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.038*** (0.014)	-1.199** (0.496)	-9.019*** (2.653)	-0.090*** (0.017)	-1.651*** (0.445)	-0.037*** (0.007)
Mean	0.00	12.27	63.09	0.00	7.50	0.065
SD	0.865	31.60	177.12	0.824	23.01	0.362
N	136699	137782	137859	136513	137874	137858

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). All regressions are based on the extended sample, which involves 19.1 weeks of the intervention period (long-term effects) as opposed to 6 weeks specified in the pre-registration (short-term effects). In addition, we exclude the top 1 percentile of observations for each outcome. The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B11: MAIN ATTRITION ANALYSIS

	Survived						Last Period					
	Hiding	Replacing	Controls	Hiding	Replacing	Controls	Hiding	Replacing	Controls	Hiding	Replacing	Controls
(Intercept)	0.768*** (0.021)	0.769*** (0.020)	0.768*** (0.021)	-0.065 (0.164)	0.115 (0.160)	-0.065 (0.164)	31.502*** (1.088)	33.252*** (0.917)	31.502*** (1.088)	-13.681 (9.088)	1.813 (7.514)	-13.681 (9.088)
GroupTreatment	-0.005 (0.030)	-0.008 (0.027)	0.000 (0.029)	0.010 (0.233)	-0.177 (0.219)	0.180 (0.229)	-0.168 (1.565)	-1.964 (1.341)	1.749 (1.423)	2.682 (12.552)	-11.561 (11.080)	15.494 (11.793)
SurvivalScore				1.099*** (0.203)	0.846*** (0.198)	1.099*** (0.203)				59.655*** (11.069)	40.697*** (9.073)	59.655*** (11.068)
GroupTreatment×SurvivalScore				-0.023 (0.289)	0.224 (0.270)	-0.253 (0.283)				-3.604 (15.240)	12.725 (13.388)	-18.959 (14.313)
N	827	983	859	777	918	807	827	983	859	777	918	807

Note: Columns 1-6 report individual-level regressions of an indicator for survival until the end of the intervention period (period 42, i.e., 42 days after the start of the intervention) on the treatment indicator. Columns titled “Hiding” pertain to the comparison between Hiding and Hide Control. Columns titled “Replacing” pertain to the comparison between Replacement and Replace Control. Columns titled “Controls” pertain to the comparison between Replace Control and Hide Control. In columns 4-6, we additionally include the survival propensity based on user characteristics—computed by regressing survival on age, male indicator, white indicator, indicator for obtaining a bachelor’s degree, indicator for being a Democrat, share of desktop usage of Facebook, active time spent on Facebook during the baseline period, content consumption on Facebook during the baseline, number of ad impressions during the baseline, and baseline number of ad clicks. Columns 7-12 report analogous regressions, with the dependent variable being the last period of recorded activity, which we use as a proxy for the last period in the study. The dependent variable is capped at 42, which indicates the last period of the six-week intervention. Note that the intervention starts in period 1, with baseline periods represented by negative numbers and zero. Standard errors clustered at the individual level are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B12: LONG-TERM ATTRITION ANALYSIS

	Hiding	Replacing	Controls	Hiding	Replacing	Controls
(Intercept)	137.751*** (5.259)	141.816*** (5.086)	137.751*** (5.258)	-50.386 (40.493)	-86.110** (42.031)	-50.386 (40.489)
GroupTreatment	0.089 (7.553)	-0.819 (6.943)	4.065 (7.316)	-11.402 (57.815)	-1.428 (56.926)	-35.724 (58.370)
SurvivalScore				248.394*** (52.183)	296.069*** (54.613)	248.394*** (52.178)
GroupTreatment×SurvivalScore				15.080 (74.458)	0.804 (73.487)	47.675 (75.544)
N	827	983	859	777	918	807

Note: Columns 1-3 report individual-level regressions of the last period of recorded activity, which we use as a proxy for the last period in the study, on the treatment indicator. Columns titled “Hiding” pertain to the comparison between Hiding and Hide Control. Columns titled “Replacing” pertain to the comparison between Replacement and Replace Control. Columns titled “Controls” pertain to the comparison between Replace Control and Hide Control. In columns 4-6, we additionally include the survival propensity based on user characteristics—computed by regressing survival on age, male indicator, white indicator, indicator for obtaining a bachelor’s degree, indicator for being a Democrat, share of desktop usage of Facebook, active time spent on Facebook during the baseline period, content consumption on Facebook during the baseline, number of ad impressions during the baseline, and baseline number of ad clicks. The dependent variable is not capped at any value. Note that the intervention starts in period 1, with baseline periods represented by negative numbers and zero. Standard errors clustered at the individual level are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B13: SHORT-TERM ENGAGEMENT EFFECTS: CLUSTERED ERRORS

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.026 (0.036)	0.646 (1.283)	5.641 (7.338)	-0.192*** (0.051)	-7.049*** (1.106)	-0.038 (0.027)
Mean	0.00	11.26	59.93	0.00	7.44	0.066
SD	0.85	28.08	175.10	0.808	25.28	0.357
N	69468	69468	69468	69468	69468	69468
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.011 (0.032)	-0.674 (1.278)	-3.738 (7.158)	-0.150* (0.082)	-1.586 (1.098)	-0.093 (0.057)
Mean	0.00	12.37	64.52	0.00	8.30	0.067
SD	0.852	31.74	204.40	0.796	32.80	0.37
N	82572	82572	82572	82572	82572	82572

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. The unit of observation is the individual-day, measured relative to the intervention date. Standard errors clustered at the individual level are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B14: LONG-TERM ENGAGEMENT EFFECTS: CLUSTERED ERRORS

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.020 (0.031)	0.572 (1.107)	6.632 (6.242)	-0.167*** (0.042)	-6.397*** (1.019)	-0.039 (0.025)
Mean	0.00	11.90	63.08	0.00	7.89	0.073
SD	0.85	30.16	180.03	0.813	26.50	0.422
N	116515	116515	116515	116515	116515	116515
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.032 (0.031)	-1.701 (1.162)	-9.401 (6.446)	-0.178* (0.093)	-2.033** (0.958)	-0.109* (0.066)
Mean	0.00	13.09	69.67	0.00	8.90	0.067
SD	0.864	33.66	203.49	0.808	30.11	0.38
N	139275	139275	139275	139275	139275	139275

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). All regressions are based on the extended sample, which involves 19.1 weeks of the intervention period (long-term effects) as opposed to 6 weeks specified in the pre-registration (short-term effects). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. The unit of observation is the individual-day, measured relative to the intervention date. Standard errors clustered at the individual level are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B15: SHORT-TERM ENGAGEMENT EFFECTS: CLUSTERED ERRORS (ASINH)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.066*	0.117*	0.202**	-0.285***	-0.687***	-0.033**
	(0.037)	(0.070)	(0.092)	(0.045)	(0.076)	(0.013)
Mean	0.00	1.20	1.63	0.00	0.904	0.05
SD	0.976	1.82	2.48	0.841	1.59	0.241
N	69468	69468	69468	69468	69468	69468
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.021	-0.043	-0.069	-0.087**	-0.048	-0.035**
	(0.032)	(0.060)	(0.081)	(0.037)	(0.050)	(0.014)
Mean	0.00	1.22	1.64	0.00	0.883	0.05
SD	0.978	1.85	2.49	0.844	1.59	0.242
N	82572	82572	82572	82572	82572	82572

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. All outcomes are transformed using the inverse hyperbolic sine (asinh) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. Standard errors clustered at the individual level are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B16: LONG-TERM ENGAGEMENT EFFECTS: CLUSTERED ERRORS (ASINH)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.054	0.103	0.175**	-0.260***	-0.640***	-0.032***
	(0.034)	(0.064)	(0.085)	(0.039)	(0.069)	(0.012)
Mean	0.00	1.22	1.72	0.00	0.951	0.053
SD	0.976	1.84	2.52	0.841	1.62	0.256
N	116515	116515	116515	116515	116515	116515
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.057*	-0.107*	-0.143*	-0.110***	-0.089*	-0.040***
	(0.031)	(0.058)	(0.078)	(0.038)	(0.051)	(0.014)
Mean	0.00	1.25	1.74	0.00	0.959	0.05
SD	0.978	1.87	2.53	0.839	1.64	0.243
N	139275	139275	139275	139275	139275	139275

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample. **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). All regressions are based on the extended sample, which involves 19.1 weeks of the intervention period (long-term effects) as opposed to 6 weeks specified in the pre-registration (short-term effects). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. All outcomes are transformed using the inverse hyperbolic sine (asinh) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. Standard errors clustered at the individual level are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

C Extension Errors

Browser extensions are complex pieces of software tailored to work with the HTML structure of specific websites. In our case, most of the functionality relates to user experience on Facebook, which the extension can seamlessly alter (via a hiding or a replacement intervention). Any change that Facebook makes to the HTML structure of its website may result in browser extension errors, invalidating parts of its functionality. This is because the extension relies on the specific structure of the website to perform its functions. In such an environment, some errors are typically inevitable. This appendix describes the errors that affected some of our participants during the study and offers robustness analysis to show that they did not impact the main conclusions of the paper.

C.1 Website Update

This error occurred as a result of a sudden change of the HTML code of the Facebook page. We are aware that it caused an issue with displaying advertiser names in replacement ads (in the replacement intervention)—the advertiser name sometimes contained wrong text. Since the HTML change potentially affected the extension’s behavior for users in all treatment groups, we assign one to the error dummy to all users in the study on the days from the beginning of the problem until the fix was implemented. The share of user-day observations in which participants experienced at least one instance of this error was 4.9%; in total, 684 users were affected at some point during the pre-registered study period.

Given that the error occurred for all people on the same days, the error dummy is subsided by period $\times$ cohort fixed effects. The error enters the regression through the dummy’s interaction with treatment. Note that we have users in our sample who already finished the pre-registered intervention period by the time that the error occurred—we use that subsample for robustness analysis.

C.2 Faulty Replacement

This error only affects some users in the replacement group. Other groups are not affected. For a period of time, the replacement intervention replaced ads not with a randomly selected ad from the pool of ads but with a default ad. This still reduced the degree of ad microtargeting but did not maintain the natural ad variety. The share of user-day observations with at least one instance of this error was 0.5%, with 133 affected users.

Since the error does not affect users in groups other than the replacement group, for the hiding regression, both the error dummy and its treatment interaction get dropped. We have users in our sample who were recruited or transitioned into the intervention period after the error was fixed—we use that subsample for robustness analysis.

C.3 Faulty Hiding

This error affected a random subset of users enrolled at the time of its occurrence. During an extension update, some users who were in the baseline period temporarily experienced the hiding intervention. This was caused by the extension overriding values specifying the extension’s behavior for a random subset of users. The share of user-day observations with at least one instance of this error was 0.4%, with 261 affected users.

Faulty hiding occurs only during the baseline period, so the error dummy’s treatment interaction gets dropped from the regressions. We have users in our sample who were recruited after the error was fixed. We also have users who were never affected, as the extension did not randomly turn on the hiding intervention for them during the baseline. All of these users constitute a subsample that we use for robustness analysis.

C.4 Robustness

For each of the three error types, we assess robustness by excluding all users affected by the respective error. All robustness exercises use specifications with an asinh-transformed outcome variable, and results are compared to the baseline asinh regressions. Because skewness in the outcome variable limits the precision of estimates in levels regressions, comparing asinh regressions across samples provides a more meaningful robustness check.

Tables B17 presents short-term results excluding individuals who experienced the website update error. All of the treatment effects reported in the original regression (Table B7) are robust to this exclusion. We do not have enough observations to perform this check for the long-term effects since all users are affected by this error unless they dropped out of the study before the error occurred.²²

Tables B18 and B19 present short-term and long-term estimates excluding individuals who experienced the faulty replacement error. Treatment effects of the hiding intervention are unaffected by this exclusion. All long-term treatment effects of the replacement inter-

²²The update causing this error happened close to the end of our study period.

vention reported in Table B8 remain robust, as do the short-term negative effects on content consumption and the ad engagement index relative to Table B7.

Lastly, Tables B20 and B21 present short-term and long-term estimates excluding participants who experienced the faulty hiding error. All short- and long-term treatment effects of the hiding intervention are robust to this exclusion. The same holds for all long-term treatment effects of the replacement intervention. The short-term negative effect of the replacement intervention on the ad engagement index also remains robust. However, we do not detect statistically significant effects of either intervention on outcomes measuring overall platform engagement.

Based on the results of our robustness analysis, we conclude that the extension errors do not meaningfully impact the treatment effects and the paper’s experimental conclusions.

TABLE B17: SHORT-TERM EFFECTS WEBSITE UPDATE ERROR (ASINH)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.058*** (0.021)	0.102*** (0.038)	0.190*** (0.054)	-0.287*** (0.015)	-0.712*** (0.032)	-0.033*** (0.005)
Mean	0.00	1.28	1.76	0.00	0.963	0.052
SD	0.974	1.85	2.55	0.839	1.62	0.245
N	41580	41580	41580	41580	41580	41580
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.022* (0.011)	-0.051** (0.025)	-0.078** (0.033)	-0.088*** (0.011)	-0.052*** (0.019)	-0.037*** (0.005)
Mean	0.00	1.31	1.80	0.00	0.959	0.055
SD	0.977	1.90	2.60	0.845	1.68	0.257
N	53004	53004	53004	53004	53004	53004

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample, excluding individuals who experienced the website update error (Section C.1). **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. All outcomes are transformed using the inverse hyperbolic sine (asinh) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B18: SHORT-TERM EFFECTS FAULTY REPLACEMENT ERROR (ASINH)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.066*** (0.021)	0.117*** (0.038)	0.202*** (0.051)	-0.285*** (0.013)	-0.687*** (0.029)	-0.033*** (0.004)
Mean	0.00	1.20	1.63	0.00	0.904	0.05
SD	0.976	1.82	2.48	0.841	1.59	0.241
N	69468	69468	69468	69468	69468	69468
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.025 (0.017)	-0.055 (0.033)	-0.085* (0.044)	-0.075*** (0.015)	-0.028 (0.028)	-0.032*** (0.004)
Mean	0.00	1.22	1.64	0.00	0.883	0.05
SD	0.978	1.85	2.49	0.844	1.59	0.242
N	71400	71400	71400	71400	71400	71400

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample, excluding individuals who experienced the faulty replacement error (Section C.2). **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. All outcomes are transformed using the inverse hyperbolic sine (asinh) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B19: LONG-TERM EFFECTS FAULTY REPLACEMENT ERROR (ASINH)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.054*** (0.017)	0.103*** (0.029)	0.175*** (0.042)	-0.260*** (0.010)	-0.640*** (0.023)	-0.032*** (0.004)
Mean	0.00	1.22	1.72	0.00	0.951	0.053
SD	0.976	1.84	2.52	0.841	1.62	0.256
N	116515	116515	116515	116515	116515	116515
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.073*** (0.016)	-0.138*** (0.032)	-0.187*** (0.040)	-0.093*** (0.015)	-0.066** (0.029)	-0.036*** (0.004)
Mean	0.00	1.25	1.74	0.00	0.959	0.05
SD	0.978	1.87	2.53	0.839	1.64	0.243
N	120088	120088	120088	120088	120088	120088

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample, excluding individuals who experienced the faulty replacement error (Section C.2). **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). All regressions are based on the extended sample, which involves 19.1 weeks of the intervention period (long-term effects) as opposed to 6 weeks specified in the pre-registration (short-term effects). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. All outcomes are transformed using the inverse hyperbolic sine (asinh) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B20: SHORT-TERM EFFECTS FAULTY HIDING ERROR (ASINH)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.072*** (0.026)	0.132*** (0.045)	0.230*** (0.063)	-0.269*** (0.017)	-0.682*** (0.032)	-0.026*** (0.005)
Mean	0.00	1.19	1.62	0.00	0.902	0.048
SD	0.977	1.81	2.48	0.84	1.60	0.237
N	59976	59976	59976	59976	59976	59976
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.007 (0.016)	-0.007 (0.033)	-0.020 (0.042)	-0.083*** (0.015)	-0.042 (0.030)	-0.034*** (0.005)
Mean	0.00	1.26	1.70	0.00	0.926	0.052
SD	0.98	1.86	2.51	0.844	1.61	0.247
N	70140	70140	70140	70140	70140	70140

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample, excluding individuals who experienced the faulty hiding error (Section C.3). **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. All outcomes are transformed using the inverse hyperbolic sine (asinh) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE B21: LONG-TERM EFFECTS FAULTY HIDING ERROR (ASINH)

	Platform Engagement			Ad Engagement		
	Index	Time	Content	Index	Impressions	Clicks
<i>Panel A: Hide vs Pure Control</i>						
Treated	0.053*** (0.019)	0.107*** (0.033)	0.188*** (0.050)	-0.249*** (0.013)	-0.632*** (0.026)	-0.026*** (0.004)
Mean	0.00	1.19	1.69	0.00	0.934	0.05
SD	0.977	1.82	2.50	0.84	1.61	0.249
N	101036	101036	101036	101036	101036	101036
<i>Panel B: Replace vs Replace Control</i>						
Treated	-0.049*** (0.014)	-0.085*** (0.029)	-0.110*** (0.036)	-0.110*** (0.016)	-0.088*** (0.028)	-0.041*** (0.005)
Mean	0.00	1.29	1.80	0.00	1.01	0.051
SD	0.98	1.89	2.56	0.837	1.67	0.245
N	118238	118238	118238	118238	118238	118238

Note: **Panel A** presents estimated treatment effects of the hiding intervention on Facebook (relative to Hide Control), based on Equation 3 and our main experimental sample, excluding individuals who experienced the faulty hiding error (Section C.3). **Panel B** reports corresponding estimates for the replacement intervention (relative to Replace Control). All regressions are based on the extended sample, which involves 19.1 weeks of the intervention period (long-term effects) as opposed to 6 weeks specified in the pre-registration (short-term effects). The dependent variables are: (1) a platform engagement index, constructed from active time spent on the platform and the number of posts and comments shown to the user; (2) active time spent on the platform; (3) number of posts and comments shown to the user; (4) an ad engagement index, constructed from the number of ad impressions and the number of ad clicks; (5) number of ad impressions; and (6) number of ad clicks. All outcomes are transformed using the inverse hyperbolic sine (asinh) function; for index variables, the transformation is applied prior to standardization. The unit of observation is the individual-day, measured relative to the intervention date. Driscoll-Kraay standard errors are reported in parentheses. All regressions were estimated using inverse probability weights reflecting the probability of group assignment for individuals recruited on a given day. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

D Recruitment and Sample

D.1 Recruitment Funnel

Our Facebook recruitment advertisements reached 175,044 unique users and generated 2,364 direct link clicks. We recorded 2,427 installations of the browser extension, and 2,279 participants completed the baseline survey.²³

D.2 Sample Selection Criteria

We preregistered the following exclusion criteria, applied to all users who installed the extension (baseline completion was not a prerequisite to participation):

- English not set as the interface language on Facebook—this may interfere with content and ad detection that were designed for and tested on English language accounts;
- Using a browser other than Chrome or Edge (both popular Chromium browsers supported by our extension)—for example, Opera or Firefox. The extension was not designed or tested for other browsers;
- Creating multiple accounts or using bots in violation of the consent form. We identified bots by flagging cases where multiple survey responses arrived simultaneously from distinct IPs with identical answers, or where extension-recorded outcomes exceeded 30 standard deviations from the mean;
- Uninstalling the extension upon installation, before the treatment period begins;²⁴
- Using Facebook for less than 1 minute per week on average during the baseline period.

In total, 617 of the 2,427 users who installed the extension are excluded from the analysis. The counts of users who failed individual exclusion criteria (which may overlap), are: insufficient Facebook activity (421), uninstallation event or failure to update the extension (239), browser other than Chrome or Edge (40), bot-like behavior (13), and unsupported language (1). After applying these criteria, we arrive at the main experimental sample of 1,810 users. Of these, 1,390 remained active at the end of the full six-week intervention period, and 818 completed the endline survey.

²³The number of installations exceeds the number of link clicks because some users returned to landing page some time after they visited it originally, others searched for the study to verify its legitimacy, etc.

²⁴We use two methods to identify cases like this. The earliest version of the extension was not able to track uninstallations, so we rely on the user never updating the extension (which happens automatically as they load the browser). Later we added functionality to track uninstallations directly.